An explicit exponent for the exceptional set of the Erdős–Straus equation

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The Erdős–Straus conjecture says that four divided by any integer greater than one is a sum of three unit fractions. In 1970 Vaughan showed that the integers up to N for which it fails number at most N times the exponential of minus a positive constant times the two-thirds power of $\log N$, with no value of the constant. We prove this bound with a constant that a computation with directed rounding shows to exceed 0.6931. The residue classes we sieve with include classes to moduli up to the three-fifths power of each prime, which two recent theorems of Maynard let us count.

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I. INTRODUCTION

For an integer $n \geq 2$ the Erdős–Straus conjecture asserts that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers x, y, z [12, §D11]. A solution for n gives one for every multiple of n, so it is enough to treat primes, and polynomial identities settle every prime outside the six residue classes $1, 121, 169, 289, 361, 529$ modulo 840 [21]. These six classes are the squares among the classes prime to 840, and Mordell observed that no identity of this kind can settle a square class.[1] The conjecture has been verified for all $n \leq 10^{17}$ by Salez [25] and for all $n \leq 10^{18}$ by Mihnea and Dumitru [18]. It remains open.

Let $\mathscr{E}(N)$ be the set of $n \leq N$ for which (1) has no solution. Vaughan [27] proved in 1970 that

$$|\mathscr{E}(N)| \ll N \exp\bigl(-c(\log N)^{2/3}\bigr) \tag{2}$$

for some constant $c > 0$, by applying the large sieve to the residue classes produced by one family of identities. Elsholtz and Tao [11] later determined the average number of solutions of (1) for prime n. Pomerance and Weingartner [23] extended (2) to m/n uniformly in m, with the exponent

[1] An identity that solves (1) for all n in a class r modulo q exists only if r is not a square modulo q; see [11, Section 2] for the principle and the families of identities behind it. The same computation decides a recent claim of a proof for the primes $p \equiv 1 \pmod 4$ [9]. It reduces the problem to a finite family of classes $p \equiv -4\alpha s^2 \pmod M$ with $M = 4\alpha sr - 1$ that is supposed to cover every prime. Since $M \equiv 3 \pmod 4$, $M \equiv -1$ modulo every odd prime factor of α, and $M \equiv -1 \pmod 8$ when α is even, quadratic reciprocity gives the Jacobi symbol $\left(\frac{-4\alpha s^2}{M}\right) = -1$, so no such class contains a square, and the class $p \equiv 1$ modulo 4 times the product of the moduli is never covered. By Dirichlet's theorem it contains infinitely many primes, all $\equiv 1 \pmod 4$. The reduction is therefore conditional, as its author states.

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$C(\log N)^{2/3}/\varphi(m)^{1/3}$ for an absolute constant C. In both papers the number of residue classes at each prime is controlled only up to constant factors, and no numerical value of c follows.[2] Dahan [7] has recently run Vaughan's large-sieve scheme on a two-parameter Type II system for the primes $n \equiv 1 \pmod{24}$, which contain every prime in $\mathscr{E}(N)$, and obtained (2) for these primes with $c = 0.065$. The sum that governs his system has the leading constant $9/(2\pi^2)$, found by Liu [14] and, with a power-saving error term, in [5]; with it the same argument gives $c = 0.0943$ for the primes. For the same primes Newton [22] studies the exact divisor tests attached to a prescribed first denominator: his finite menu of Type II congruences covers the proportion $25651/26496$ of their reduced residue classes, and he shows that several ways of combining the tests at different denominators cannot by themselves force a solution.[3] Here we give an explicit constant for the whole set $\mathscr{E}(N)$, and we take it beyond the value that the residue classes reachable by the Bombieri–Vinogradov theorem can give.

Theorem A. *For every $\varepsilon > 0$ there is $N_0(\varepsilon)$ such that, for all $N \ge N_0(\varepsilon)$,*

$$|\mathscr{E}(N)| \le N \exp\bigl(-(c_1 - \varepsilon)(\log N)^{2/3}\bigr), \tag{3}$$

where

$$c_1 = \frac{3}{2}\Bigl(\frac{9}{8\pi^2}\Bigl(\frac{40}{49} - 6\mathscr{J} + 6\mathscr{K}\Bigr)\Bigr)^{1/3}, \tag{4}$$

$\mathscr{J}$ is the integral (30) and $\mathscr{K}$ the measure (29) of Section VII. Moreover $0 \le \mathscr{J} < 0.000335$ and $\mathscr{K} > 0.00859$, so that $c_1 > 0.6931$; in closed form $\mathscr{J} \le 533/64680$ and $\mathscr{K} \ge 0$, so that

$$c_1 \ge c_{\mathrm{var}} = \Bigl(\frac{286983}{98560\,\pi^2}\Bigr)^{1/3} = 0.6657102497\ldots \tag{5}$$

The number $40/49$ is the share of the triangle of factorisations (Figure 1) covered by the corners and by the strips of the centre along them, $6\mathscr{J}$ is the part of that share which our count of the strips misses, and $6\mathscr{K}$ is the share added by the deep strips behind them, which we count through a second theorem of Maynard. Numerical integration gives $\mathscr{J} = 0.0003$ (Section X), and two independent programs and a Monte Carlo simulation agree with it. A second simulation gives $\mathscr{K} \approx 0.0138$ and $c_1 \approx 0.7015$. The bounds $\mathscr{J} < 0.000335$ and $\mathscr{K} > 0.00859$ are proved by computations in which every rounding error is directed against the bound (Propositions X.2 and X.3), and the weaker bound (5) in closed form.

[2] In [23, Section 4] the number of classes, summed over the primes $p \le x$ with weight $1/p$, is shown to be of order $(\log x)^2$: the upper bound comes from the Brun–Titchmarsh inequality and the lower bound from the Bombieri–Vinogradov theorem, with different implied constants, and the exponent is written through these two constants.

[3] Each family of [22, Section 3] is a residue class of n to a fixed modulus $4j-1$ with $j \le 12$. Such classes settle a fixed proportion of the primes up to N and leave the exponent in (2) unchanged. The large sieve used here takes classes of n modulo primes p that grow with N, and the constant comes from the number of these classes at each p.

The second theorem concerns the classes that the Bombieri–Vinogradov theorem can count. For a prime $p \equiv 3 \pmod 4$ with $P = (p+1)/4$, let $\Omega_p^\sharp$ be the set of classes $-u/a \pmod p$ attached to factorisations $P = ua\alpha$ with $\gcd(u,a) = 1$ in which one of the three products ua, $u\alpha$, $a\alpha$ is at most $\sqrt{p}$ (Section II). Let $L^\sharp(Q)$ be the sum that the large sieve attaches to these classes, as in (31) below.

Theorem B. *For every $\varepsilon > 0$ there is $N_1(\varepsilon)$ such that, for all $N \ge N_1(\varepsilon)$ and all $Q \ge 1$,*

$$\frac{N+Q^2}{L^\sharp(Q)} \ge N \exp\bigl(-(c_0 + \varepsilon)(\log N)^{2/3}\bigr),$$

where

$$c_0 = \frac{9}{2^{8/3}\pi^{2/3}} = \frac{3}{2}\Bigl(\frac{27}{32\pi^2}\Bigr)^{1/3} = 0.6607886973\ldots \tag{6}$$

The classes $\Omega_p^\sharp$ come from the three corners of the triangle, and with them the large sieve gives exactly the constant c_0: Theorem B is the upper bound, and the corners alone already give c_0 (Corollary V.3). The number $27/(32\pi^2)$ inside (6) is the density of these classes. The gain from c_0 to c_1 comes from the classes of the centre of the triangle, which lie beyond the reach of the Bombieri–Vinogradov theorem.

The method. Each prime $p \equiv 3 \pmod 4$ carries the identities of Type II [11]: for every factorisation $P = ua\alpha$ with $\gcd(u,a) = 1$, every $n \equiv -u/a \pmod p$ has a solution of (1) (Lemma II.2). In logarithmic coordinates these factorisations fill a triangle (Figure 1). To sieve with their classes one must count them over the primes and show that they are distinct. Counting requires primes in progressions to the moduli $4ua$, $4u\alpha$ or $4a\alpha$. The Bombieri–Vinogradov theorem controls such moduli up to about $\sqrt{p}$, which cuts out three corners of the triangle, where one product of two variables is at most $\sqrt{p}$. In each corner the class has a normal form of small height, u/a, $4u^2\alpha$ or $(4a^2\alpha)^{-1}$, which separates the classes inside that corner, and two arguments with explicit inequalities show that the corner A shares no class with B or C. The corners B and C do share classes, but only along a family described by six integers, which contributes a convergent series (Proposition III.2). The corners cover three quarters of the triangle and give c_0.

The rest of the triangle is the centre, where u, a and α are all below $\sqrt{P}$. There the class u/a is a Farey fraction of order $\sqrt{P}$, so the classes of the centre are distinct. They differ from the classes of the corners, apart from a family described by five integers, which contains one explicit polynomial family of primes and again contributes a convergent series (Propositions IV.3 and IV.4). To count the centre we use a theorem of Maynard [16]: the primes $p \equiv -1 \pmod q$ have the expected number for almost all moduli q up to $p^{1/2+\vartheta}$ with $\vartheta < 1/42$, provided that q has a divisor in a range that depends on ϑ. A divisor chain (Lemma VI.3) finds such a divisor for most moduli in three strips of the centre along the corners, and we compute the proportion it misses (Lemma VI.4). If nothing were missed, the strips would add $6(\frac{1}{84} - \frac{3}{3528}) = \frac{13}{196}$ to the share $\frac{3}{4}$ of the corners; the loss $6\mathscr{J}$ is 0.0018.

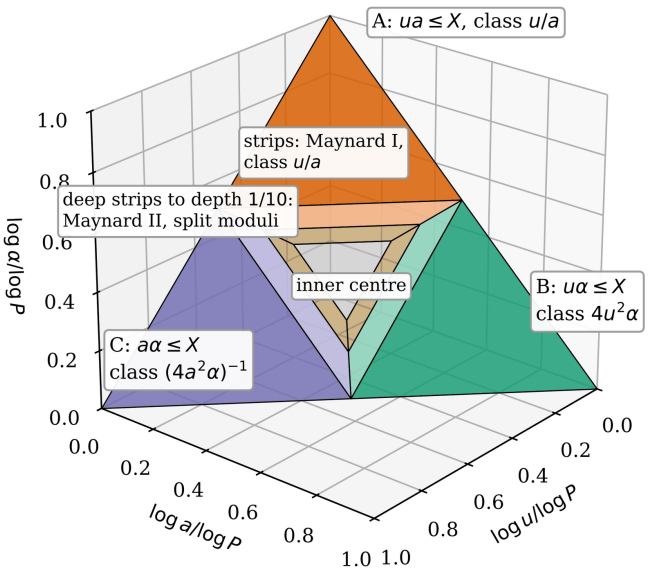


FIG. 1. In the coordinates $(\log u, \log a, \log\alpha)/\log P$ the factorisations $P = ua\alpha$ fill a triangle. The three corners A, B, C are the parts in which one product of two variables is at most X; each carries its own normal form of the class u/a (Lemma II.3). At $X = \sqrt{p}$ each corner has a quarter of the area. In the centre the three coordinates are below $\frac{1}{2}$ and the class u/a itself separates the classes (Proposition IV.3). The light strips of the centre along the corners, of depth $\frac{1}{42}$, are counted with Maynard's theorem (Section VI); they are drawn wider than they are. The deep strips behind them, to the depth $\frac{1}{10}$, are counted in part with Maynard's second theorem (Section VII). The inner centre would need moduli beyond $p^{3/5}$.

Beyond the depth $\frac{1}{42}$ the range of divisors in Maynard's theorem is empty. There we use his second theorem [17], which counts primes in progressions to moduli beyond $p^{1/2}$ when the weight attached to the moduli is a convolution of three sequences of suitable sizes. We prove a form of it in which the sizes of the three factors may depend on the size of the long variable of Heath-Brown's identity (Theorem VII.1), and we split each modulus according to the sizes of its prime factors: the large prime factors are placed in three groups and the small ones fill two of the groups to prescribed sizes (Definition VII.2 and Figure 3). The pairs whose moduli split in this way have measure $\mathscr{K}$ in the rest of the centre, the deep strips, and add $6\mathscr{K}$ to the share. No splitting exists beyond the depth $\frac{1}{10}$ (Lemma VII.4). A computation that bounds the density of the factorable pairs from below on cells of depth (Proposition X.3) proves $\mathscr{K} > 0.00859$.

The corners, their normal forms, the treatment of the centre, the rule that finds the divisors and the splitting of the moduli in the deep strips were selected by the engine Anaxia of the BLLM project [1, 3].[4]

The rest of the argument is analytic. A Rankin-type upper bound and a tilted lower bound determine the sum of the large sieve to leading order once the density of the classes is known (Proposition VIII.2); the same upper bound proves Theorem B. Section X computes the constant, Section XI compares regions of the triangle (Table II) and explains where the method stops, and Appendix A contains every computation in full.

II. TYPE II TRIPLES AND THEIR CLASSES

Throughout, p denotes a prime with $p \equiv 3 \pmod 4$, and $P = (p+1)/4$, so that $p = 4P-1$ and $4P \equiv 1 \pmod p$.

Definition II.1. A *Type II triple* for p is $t = (u,a,\alpha) \in \mathbb{N}^3$ with $ua\alpha = P$ and $\gcd(u,a) = 1$. Their set is $\mathscr{T}(p)$. The *class* of t is $\mathrm{c}(t) = ua^{-1} \bmod p$, which is defined because $1 \le a \le P < p$.

Lemma II.2. *Let $t = (u,a,\alpha) \in \mathscr{T}(p)$ and $n \ge 1$ with $n \equiv -\mathrm{c}(t) \pmod p$. Then $v = (an+u)/p$ is a positive integer and*

$$\frac{4}{n} = \frac{1}{\alpha uv} + \frac{1}{a\alpha nu} + \frac{1}{a\alpha nv}. \tag{7}$$

Proof. The congruence $n \equiv -u/a$ means $p \mid an+u$, and $an+u \ge 2$, so v is a positive integer. Then $an+u = pv = (4ua\alpha - 1)v$, that is $an+u+v = 4a\alpha uv$. Dividing by $a\alpha nuv$ gives (7). □

When n is prime and $n \nmid uv$, two of the three denominators in (7) are divisible by n and the third is not; these are the solutions of Type II in the sense of [11].[5] By Lemma II.2, an integer $n \in \mathscr{E}(N)$ lies, for every prime $p \equiv 3 \pmod 4$, outside every class $-\mathrm{c}(t)$ with $t \in \mathscr{T}(p)$. This is the input to the large sieve.

Lemma II.3. *For every $t = (u,a,\alpha) \in \mathscr{T}(p)$,*

$$\mathrm{c}(t) \equiv 4u^2\alpha \equiv (4a^2\alpha)^{-1} \pmod p. \tag{8}$$

Proof. From $p = 4ua\alpha - 1$ we get the identities

$$4(u^2\alpha)a - u = up, \qquad 4(a^2\alpha)u - a = ap.$$

The first gives $4u^2\alpha \cdot a \equiv u$, so $u/a \equiv 4u^2\alpha$. The second gives $4a^2\alpha \cdot u \equiv a$, so $u/a \equiv (4a^2\alpha)^{-1}$. □

[4] Anaxia builds constructions by typed crossing of proof components and tests them on data. We ran version 3.0.5 twice and version 3.1.0 once. On the corners it tested 128 constructions on the 39,322 primes $p \equiv 3 \pmod 4$ below 10^6 and kept the seven built from one, two or three corners with matching normal forms; the best of them gives c_0. On the centre it tested 37 constructions and accepted five, the best of which is the construction of this paper. It rejected the forms $4u^2\alpha$ and $(4a^2\alpha)^{-1}$ in the centre, and a rule that omits the bound on the steps of the divisor chain, by finding counterexamples on data. On the deep strips, version 3.1.0 [2] tested 15 constructions: it rejected Maynard's window beyond the depth $\frac{1}{42}$, where it is empty, and every rule that fills two groups without a reserve for the overshoot of the first, by finding profiles that the fill cannot bring to its targets; the best of the constructions it accepted splits the moduli as in Definition VII.2. Section XII describes the three runs in full. The engine selected the constructions; the proofs are ours.

[5] The identity (7) with $p = 4ua\alpha - 1$ is the one behind Vaughan's argument [27] and the Type II search of [7]; in [5] we used it with α replaced by the witness s of that search.

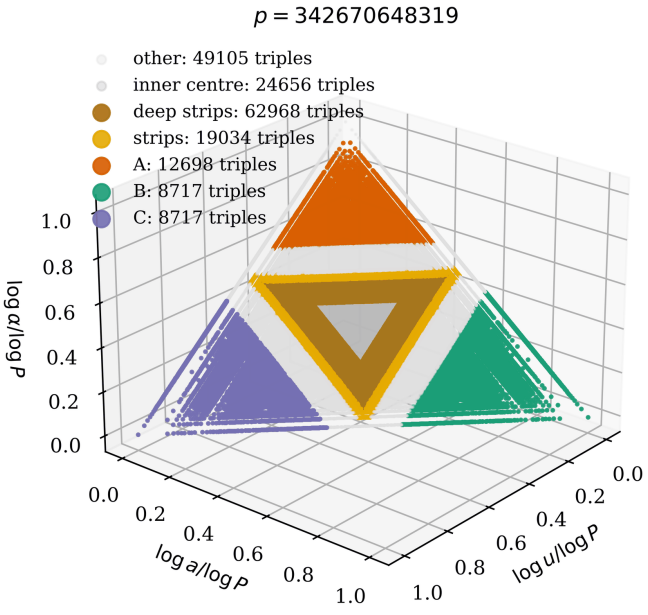


FIG. 2. The 185,895 Type II triples of the prime $p = 342670648319$, for which $P = 2^8 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23$, with the corners at level $X = \sqrt{p}/16$, the strips of the centre (largest coordinate at least $P^{1/2-1/42}$), the deep strips (largest coordinate from $P^{2/5}$ to $P^{1/2-1/42}$) and the inner centre. The light grey triples lie in no corner and outside the centre. The corners B and C have the same size, as the involution ι shows.

We call u/a, $4u^2\alpha$ and $(4a^2\alpha)^{-1}$ the three normal forms of the class. Each is useful where its height is small.

Definition II.4. For $X \geq 1$ the three *corners* at level X are

$$A_p(X) = \{t \in \mathscr{T}(p) : u \geq 2,\ a \geq 2,\ ua \leq X\},$$
$$B_p(X) = \{t \in \mathscr{T}(p) : u\alpha \leq X\},$$
$$C_p(X) = \{t \in \mathscr{T}(p) : a\alpha \leq X\}.$$

The corner A omits the triples with $u = 1$ or $a = 1$, which are needed in the proof of Proposition III.1 and are too few to matter in the count (they contribute $O(\log X)$ to the sums of Section V). The map $\iota(u,a,\alpha) = (a,u,\alpha)$ is an involution of $\mathscr{T}(p)$ with $\mathrm{c}(\iota t) = \mathrm{c}(t)^{-1}$; it maps $A_p(X)$ onto itself and exchanges $B_p(X)$ and $C_p(X)$. In particular $|B_p(X)| = |C_p(X)|$.

Lemma II.5. *Suppose $16X^2 \leq p$. Then:*

(i) the corners $A_p(X)$, $B_p(X)$, $C_p(X)$ are pairwise disjoint;

(ii) on $A_p(X)$ one has $u,a \leq X$; on $B_p(X)$ one has $u^2\alpha \leq X^2$; on $C_p(X)$ one has $a^2\alpha \leq X^2$.

Proof. (i) Two of the three conditions force the product of the two products to be at most X^2. For A and B this is $(ua)(u\alpha) = uP \leq X^2$; for A and C it is $aP \leq X^2$; for B and C it is $\alpha P \leq X^2$. In each case $P \leq X^2 \leq p/16 < P$, which is impossible. (ii) On A, u and a divide $ua \leq X$. On B, $u^2\alpha = u \cdot u\alpha \leq (u\alpha)^2 \leq X^2$, and similarly on C. □

III. DISTINCT CLASSES IN THE CORNERS

This section proves that the classes of the three corners are distinct, apart from a sparse set of coincidences between B and C.

Proposition III.1. *Suppose $16X^2 \leq p$. Then* c *is injective on each of $A_p(X)$, $B_p(X)$ and $C_p(X)$, and no class of $A_p(X)$ is a class of $B_p(X)$ or of $C_p(X)$.*

Proof. Inside A. Let $t,t' \in A_p(X)$ with $u/a \equiv u'/a'$. Then $p \mid ua' - u'a$, and $|ua' - u'a| < X^2 < p$ by Lemma II.5(ii), so $ua' = u'a$. Since $\gcd(u,a) = \gcd(u',a') = 1$, this gives $u = u'$, $a = a'$ and then $\alpha = \alpha'$.

Inside B. Let $t,t' \in B_p(X)$ with equal classes. By (8), $4u^2\alpha \equiv 4u'^2\alpha'$, and both sides lie in $[1,4X^2] \subset [1,p)$, so $h = u^2\alpha$ equals $h' = u'^2\alpha'$. Now $ha = u \cdot ua\alpha = uP$, so $u/a = h/P$ as rational numbers; since $\gcd(u,a) = 1$, the pair (u,a) is the reduced form of h/P. The same holds for t', so $t = t'$.

Inside C. The involution ι maps C onto B and inverts classes, so injectivity on C follows from injectivity on B.

A against B. Let $t = (u,a,\alpha) \in A_p(X)$ and $t' = (u',a',\alpha') \in B_p(X)$ have the same class, and put $h = u'^2\alpha'$, so that $u/a \equiv 4h$ by (8). Then $4ah - u = wp$ for some integer w.

If $w \leq -1$ then $4ah \leq u - p < 0$, which is impossible. If $w = 0$ then $4ah = u$, so $a \mid u$, and $\gcd(u,a) = 1$ forces $a = 1$, which is excluded in A. Hence $w \geq 1$. Substituting $p = 4ua\alpha - 1$ in $4ah = u + wp$ gives

$$4ah = u - w + 4ua\alpha w,$$

so $4a \mid u - w$. Write $u - w = 4aj$ with $j \in \mathbb{Z}$; then

$$h = j + wu\alpha, \qquad w = u - 4aj. \tag{9}$$

Case $j \leq -1$. Then $w = u + 4a|j|$ and

$$h = -|j| + (u + 4a|j|)u\alpha = |j|p + u^2\alpha > p,$$

while $h \leq X^2 < p$ by Lemma II.5(ii). This is impossible.

Case $j = 0$. Then $h = u^2\alpha$, and, as in the proof for B, $u'/a' = h/P = u^2\alpha/(ua\alpha) = u/a$, so $t' = t$. But A and B are disjoint by Lemma II.5(i).

Case $j \geq 1$. Since $w \geq 1$, (9) gives $4aj \leq u - 1$, so $aj < u/4$. Put $d = u'\alpha'$. Then $P = a'd$ and $h = u'd$, and from (9),

$$ah = aj + wua\alpha = aj + wP = aj + wa'd,$$

that is

$$d\,(au' - a'w) = aj. \tag{10}$$

The right side is positive, so $au' - a'w \geq 1$ and $d \leq aj < u/4$. Hence $h = u'^2\alpha' \leq (u'\alpha')^2 = d^2 < u^2/16$, and

$$p < u + wp = 4ah < \frac{au^2}{4} \leq \frac{(ua)^2}{4} \leq \frac{X^2}{4} < p,$$

a contradiction. So no class of A is a class of B.

A against C. If $t \in A_p(X)$ and $t' \in C_p(X)$ had the same class, then $\iota t \in A_p(X)$ and $\iota t' \in B_p(X)$ would have the same class $\mathrm{c}(t)^{-1}$, which the previous step excludes. □

The corners B and C can share a class. The next proposition describes every such coincidence.

Proposition III.2. *Suppose $16X^2 \le p$, and let $t=(u,a,\alpha) \in B_p(X)$ and $t'=(u',a',\alpha') \in C_p(X)$ have the same class. Put*

$$d=u\alpha, \quad e=a'\alpha', \quad M=ua', \quad k=\alpha\alpha',$$

$G=\gcd(d,e)$, $d=Gd_1$ and $e=Ge_1$. Then there are integers $r,m,\tau \ge 1$ such that

$$P = Gd_1e_1(4Gr+\tau), \tag{11}$$

$$M = d_1e_1\tau(4Gr+\tau)+r, \tag{12}$$

$$G = 4k\tau r+m, \tag{13}$$

$$m(4k\tau r+m) = k\tau^2 + \frac{kr}{d_1e_1}, \quad d_1e_1 \mid kr, \tag{14}$$

and

$$\tau \ge 4mr, \qquad G \ge 16kmr^2, \qquad P \ge 1024k^2m^2r^5d_1e_1. \tag{15}$$

The integers k,m,r,d_1,e_1,τ and u determine the pair (t,t').

Proof. Step 1: the integer r. By (8), $4u^2\alpha \equiv (4a'^2\alpha')^{-1}$, so $16u^2\alpha a'^2\alpha' \equiv 1 \equiv 16P^2$ and therefore $u^2\alpha a'^2\alpha' \equiv P^2$, that is, $deM \equiv P^2 \pmod p$. On the other hand $P=ad=u'e$, so $P^2=de\cdot au'$. Since $p \nmid de$, we get $au' \equiv M \pmod p$; write $au'=M+rp$ with $r\in\mathbb{Z}$, so that

$$P^2 = de\,(M+rp). \tag{16}$$

If $r=0$ then $au'=ua'$, and since both pairs are coprime, $u=u'$ and $a=a'$, so $t=t'$, which contradicts Lemma II.5(i). If $r\le -1$ then $M=au'+|r|p>p$, but $M=ua' \le (u\alpha)(a'\alpha') \le X^2 < p$. Hence $r\ge 1$.

Step 2: the integer $s=P-4der$. Expanding (16) with $p=4P-1$ gives

$$Ps = de\,(M-r), \qquad s = P-4der. \tag{17}$$

Since $M\ge 1$, (16) gives $P^2 > der(4P-1)$, hence $4der < P^2/(P-\frac14) < P+1$, so $s\ge 0$. If $s=0$ then $M=r$ and $P=4der$; then $a=P/d=4er=4eua'$, so $u\mid a$ and $u=1$; similarly $u'=P/e=4dr=4dua'$, so $a'\mid u'$ and $a'=1$. Then $r=M=1$ and $P=4de\le 4X^2 \le p/4 < P$, a contradiction. Hence $s\ge 1$.

Step 3: the parametrisation. Both d and e divide P and $4der$, hence s, so the least common multiple Gd_1e_1 divides s. Write $s=Gd_1e_1\tau$ with $\tau\ge 1$. Then $P=s+4der=Gd_1e_1(4Gr+\tau)$, which is (11). Inserting this into (17) and cancelling $G^2d_1e_1$ gives (12). Next, $de=ua'\cdot\alpha\alpha'=kM$, so $kM=G^2d_1e_1$. With (12) this reads $kd_1e_1\tau(4Gr+\tau)+kr=G^2d_1e_1$, so $d_1e_1\mid kr$ and

$$G^2 - k\tau(4Gr+\tau) = \frac{kr}{d_1e_1} \ge 1. \tag{18}$$

Hence $G(G-4k\tau r) > k\tau^2 > 0$, so $m=G-4k\tau r\ge 1$, which is (13), and (18) becomes (14).

Step 4: the inequalities. From (14), $4k\tau rm+m^2 \le k\tau^2+kr$. Suppose $\tau < 4mr$. The integers τ and $4mr-\tau$ are positive with sum $4mr$, so their product is at least $4mr-1 \ge 3r \ge r$. Then $k\tau(4mr-\tau) \ge kr$, while the inequality above gives $k\tau(4mr-\tau) \le kr-m^2 < kr$. This contradiction proves $\tau \ge 4mr$. Then $G=4k\tau r+m \ge 16kmr^2$, and from (11),

$$P \ge Gd_1e_1\cdot 4Gr = 4rG^2d_1e_1 \ge 1024k^2m^2r^5d_1e_1.$$

Step 5: recovery of the pair. Given k,m,r,d_1,e_1,τ, the integers G, P, d, e and M are fixed by (13), (11) and (12). Given also u, we recover $\alpha=d/u$, $a'=M/u$, $\alpha'=e/a'$, $a=P/d$ and $u'=P/e$. □

For a prime $p\equiv 3 \pmod 4$ let $\nu(p)$ be the number of pairs (t,t') with $t\in B_p(\sqrt{p}/4)$, $t'\in C_p(\sqrt{p}/4)$ and $\mathrm{c}(t)=\mathrm{c}(t')$. For every $X\le \sqrt{p}/4$ the corners at level X lie inside those at level $\sqrt{p}/4$, so the number of coincidences between $B_p(X)$ and $C_p(X)$ is at most $\nu(p)$.

Corollary III.3. $\sum_p \nu(p)/p < \infty$.

Proof. Fix $\eta\in(0,\frac14)$. By Proposition III.2, each coincidence at p gives integers $k,m,r,d_1,e_1,\tau\ge 1$ that fix P, together with a divisor u of $M<p$; there are at most $C_\eta P^\eta$ such divisors. For fixed k,m,r,d_1,e_1, the equation (14) is a quadratic equation in τ with leading coefficient $k\ne 0$, so it has at most two solutions. The pairs (d_1,e_1) with $d_1e_1\mid kr$ number at most $\tau_3(kr)\le C_\eta(kr)^\eta$, where τ_3 counts ordered factorisations into three factors. Using $p>P\ge 1024k^2m^2r^5$, we obtain

$$\sum_p \frac{\nu(p)}{p} \le C'_\eta \sum_{k,m,r\ge 1} \frac{(kr)^\eta}{(k^2m^2r^5)^{1-\eta}},$$

and the series converges because the exponents of k, m and r are $2-3\eta$, $2-2\eta$ and $5-6\eta$, all greater than 1. □

IV. THE CENTRE

Definition IV.1. The *centre* of $\mathscr{T}(p)$ is

$$\mathscr{Z}_p = \{t\in\mathscr{T}(p) : u^2<P,\ a^2<P,\ \alpha^2<P\}.$$

Lemma IV.2. *Let $t=(u,a,\alpha)\in\mathscr{Z}_p$. Then $u,a,\alpha\ge 2$, $u\ne a$, and each of the products ua, $u\alpha$, $a\alpha$ exceeds $\sqrt{P}$. In particular $\mathscr{Z}_p$ is disjoint from the corners at every level $X\le\sqrt{p}/2$, and ι maps $\mathscr{Z}_p$ onto itself.*

Proof. Since $a\alpha < \sqrt{P}\sqrt{P}=P$, we have $u=P/(a\alpha)>1$, and in the same way $a,\alpha\ge 2$. Then $u\ne a$ because $\gcd(u,a)=1$. Next $ua=P/\alpha > P/\sqrt{P}=\sqrt{P}$, and in the same way for the other products. Finally $\sqrt{P}>\sqrt{p}/2$, and the conditions of the centre are symmetric in u and a. □

In the centre the first normal form u/a has numerator and denominator below $\sqrt{P}$, as for a Farey fraction of order $\sqrt{P}$, and this separates the classes.

Proposition IV.3. *Let $X\le\sqrt{p}/4$.*

(i) c *is injective on $\mathscr{Z}_p$, and no class of $\mathscr{Z}_p$ is a class of $A_p(X)$.*

(ii) No class of a triple of $\mathscr{Z}_p$ with $u<a$ is a class of $B_p(X)$, and no class of a triple of $\mathscr{Z}_p$ with $u>a$ is a class of $C_p(X)$.

Proof. (i) Let $t,t'\in\mathscr{Z}_p$ have the same class, so that $p\mid ua'-u'a$. From $u^2<P$ and $a'^2<P$ we get $(ua')^2<P^2$, so $ua'<P$, and in the same way $u'a<P$. Hence $|ua'-u'a|<P<p$, so $ua'=u'a$, and as before $t=t'$. If instead $t'\in A_p(X)$, then $u',a'\le X$ by Lemma II.5(ii), so ua' and $u'a$ are below $\sqrt{P}\cdot\sqrt{p}/4<p$; again $ua'=u'a$ and $t=t'$, which contradicts Lemma IV.2.

(ii) Let $t\in\mathscr{Z}_p$ with $u<a$. Then $4u^2\alpha<4ua\alpha=p+1$, and $4u^2\alpha\neq p$ because p is odd, so $1\le 4u^2\alpha<p$. If $t'\in B_p(X)$ has the same class, then $1\le 4u'^2\alpha'\le 4X^2<p$ by Lemma II.5(ii), and (8) gives $4u^2\alpha=4u'^2\alpha'$. As in the proof of Proposition III.1, both (u,a) and (u',a') are the reduced form of $u^2\alpha/P$, so $t=t'$, which contradicts Lemma IV.2. The second statement follows by applying ι, which maps the triples of $\mathscr{Z}_p$ with $u>a$ to those with $u<a$, maps $C_p(X)$ onto $B_p(X)$, and inverts classes. □

The remaining coincidences, between the centre with $u<a$ and the corner C (and their images under ι), have a parametrisation by five integers.

Proposition IV.4. *Let $X\le\sqrt{p}/4$, and let $t=(u,a,\alpha)\in\mathscr{Z}_p$ with $u<a$ and $t''=(u'',a'',\alpha'')\in C_p(X)$ have the same class. Put $s=a''\alpha''$. Then there are integers $k,D,j\ge 1$ such that*

$$u''a-ua''=kp, \tag{19}$$

$$a=j+4ks, \tag{20}$$

$$uD=sk, \tag{21}$$

$$\alpha j(j+4ks)=sa''-D, \tag{22}$$

and $4ks<a<\sqrt{P}$. The integers a'', α'', k, D and j determine the pair (t,t'').

Proof. The equality of classes gives $p\mid u''a-ua''$; write $u''a-ua''=kp$. If $k=0$ then $u''a=ua''$ and $t=t''$, which contradicts Lemma IV.2. If $k\le -1$ then $ua''\ge p+u''a>p$, while $ua''<\sqrt{P}\cdot X<p$. Hence $k\ge 1$. Multiplying (19) by s and using $u''s=P$ and $p=4P-1$ gives

$$P(a-4ks)=s(ua''-k).$$

From $k(4P-1)<u''a$ we get $ks(4P-1)<Pa<P(4P-1)$, so $ks<P$. If $4ks>a$, the left side is at most $-P<-sk\le s(ua''-k)$, which is impossible. So $j=a-4ks\ge 0$. If $j=0$ then $ua''=k$, so $a=4ks=4ua''s$ is a multiple of u, and $\gcd(u,a)=1$ forces $u=1$, which contradicts Lemma IV.2. Hence $j\ge 1$, which is (20). Inserting $P=ua\alpha$,

$$ua\alpha j=s(ua''-k),\qquad\text{so}\qquad u(sa''-a\alpha j)=sk.$$

Put $D=sa''-a\alpha j$; then $uD=sk\ge 1$, so $D\ge 1$, which is (21), and $\alpha ja=sa''-D$ is (22). Finally $a<\sqrt{P}$ because $t\in\mathscr{Z}_p$, and $4ks=a-j<a$. Given a'', α'', k, D and j, we recover $s=a''\alpha''$, $u=sk/D$, $a=j+4ks$, $\alpha=(sa''-D)/(ja)$, $P=ua\alpha$ and $u''=P/s$. □

For a prime $p\equiv 3\pmod 4$ let $\nu'(p)$ be the number of pairs (t,t'') with $t\in\mathscr{Z}_p$, $t''\in B_p(\sqrt{p}/4)\cup C_p(\sqrt{p}/4)$ and $\mathrm{c}(t)=\mathrm{c}(t'')$.

Corollary IV.5. $\sum_p \nu'(p)/p<\infty$.

Proof. By Proposition IV.3(ii), a pair counted by $\nu'(p)$ has either $u<a$ and $t''\in C_p(\sqrt{p}/4)$, or $u>a$ and $t''\in B_p(\sqrt{p}/4)$, and ι exchanges the two kinds. It is therefore enough to count the first kind. By Proposition IV.4, a pair of the first kind at a prime $p\le y$ gives $s=a''\alpha''$ and k with $ks<\sqrt{P}/4<\sqrt{y}$, a divisor D of sk, and a divisor j of $sa''-D$, which lies in $[1,s^2]\subset[1,y]$. Fix $\eta>0$. For given (s,k) there are at most $\tau(s)\tau(sk)\max_{m\le y}\tau(m)\ll_\eta y^\eta$ choices of (a'',α'',D,j), and the pairs (s,k) with $ks<\sqrt{y}$ number $O(\sqrt{y}\log y)$. Hence the pairs of the first kind at the primes $p\le y$ number $O_\eta(y^{1/2+2\eta})$. With $\eta=\frac18$, partial summation gives $\sum_{p\le y}\nu'(p)/p=O(1)$. □

Remark IV.6 (A family of coincidences)**.** For an integer $\alpha\ge 1$ put

$$s=16\alpha+1,\quad a=8(8\alpha+1),\quad P=sa\alpha,\quad p=4P-1.$$

When p is prime, $t=(s,a,\alpha)$ and $t''=(a\alpha,s,1)$ are Type II triples for p with the same class, since

$$(a\alpha)a-s\cdot s=4P-1=p.$$

They are coprime pairs because s is odd with $2(8\alpha+1)-s=1$, and $\gcd(s,\alpha)=1$. This is the case $(k,D,j)=(1,1,4)$ of Proposition IV.4, with $a''=s$ and $\alpha''=1$. The triple t lies in the centre exactly when $a^2<P$, that is $16\alpha^2-63\alpha-8>0$, that is $\alpha\ge 5$, and t'' lies in $C_p(\sqrt{p}/4)$ because $s<16\alpha^{3/2}<\sqrt{p}/4$. The value $\alpha=5$ gives the prime $p=531359$, with $t=(81,328,5)$ and $t''=(1640,81,1)$:

$$81^2-328\cdot 1640=-531359.$$

The value $\alpha=7$ gives the prime $p=1442783$, with $t=(113,456,7)$ and $t''=(3192,113,1)$. Together with their images under ι, these four pairs are all the pairs counted by $\nu'(p)$ for the primes $p\equiv 3\pmod 4$ below 10^7. Since p is a cubic polynomial in α, the family by itself contributes a convergent series to $\sum_p\nu'(p)/p$.

V. COUNTING THE CLASSES OF THE CORNERS

A. Mean values

Let $\kappa=9/(2\pi^2)=0.4559453\ldots$, the constant of [5, 14]. For $X\ge 1$ define

$$S_A(X)=\sum_{\substack{u,a\ge 2,\ ua\le X\\ \gcd(u,a)=1}}\frac{1}{\varphi(4ua)},$$

$$S_B(X)=\sum_{u\alpha\le X}\frac{\varphi(u)}{u\,\varphi(4u\alpha)}.$$

The weight $1/\varphi(4ua)$ is the density of the primes $p \equiv -1 \pmod{4ua}$, that is, of the primes for which (u,a,α) with $\alpha = P/(ua)$ is a triple of $A_p(X)$. In S_B the extra factor $\varphi(u)/u$ is the density of the condition $\gcd(u,a)=1$ among the primes $p \equiv -1 \pmod{4u\alpha}$.[6]

Lemma V.1. *As $X \to \infty$,*

$$S_A(X) = \frac{\kappa}{2}\log^2 X + O(\log X),$$
$$S_B(X) = \frac{\kappa}{2}\log^2 X + O(\log X).$$

Sketch. Since $\varphi(4n) = 2\varphi(n)$ for odd n and $4\varphi(n)$ for even n, we have $1/\varphi(4n) = \psi(n)/(2n)$ with $\psi(n) = \prod_{q\mid n,\, q \text{ odd}} q/(q-1)$. Thus $S_B(X) = \frac{1}{2}\sum_{u\alpha \le X} W_B(u,\alpha)/(u\alpha)$ with $W_B(u,\alpha) = \psi(u\alpha)\varphi(u)/u$, a function that is multiplicative in the pair. Writing $W_B(u,\alpha) = \sum_{d\mid u, e\mid \alpha} g(d,e)$, the function g is supported on $g(1,1)=1$, $g(1,q) = 1/(q-1)$ and $g(q,q) = -1/(q-1)$ for odd primes q, and $g(2,1) = -\frac{1}{2}$. Then

$$S_B(X) = \frac{1}{2}\sum_{d,e} \frac{g(d,e)}{de} H\Big(\frac{X}{de}\Big), \qquad H(Y) = \sum_{mn \le Y} \frac{1}{mn},$$

and $H(Y) = \frac{1}{2}\log^2 Y + 2\gamma \log Y + O(1)$ for $Y \ge 1$. The sum $\sum_{d,e} g(d,e)/(de)$ is the Euler product of the local factors $\frac{3}{4}$ at 2 and $1+q^{-2}$ at odd q, that is $\frac{3}{4}\cdot\frac{12}{\pi^2} = \frac{9}{\pi^2} = 2\kappa$, and the main term is $\frac{1}{2}\cdot 2\kappa \cdot \frac{1}{2}\log^2 X$. The sum S_A is treated in the same way with $W_A(u,a) = \psi(ua)\mathbf{1}[\gcd(u,a)=1]$; its local factors are again $\frac{3}{4}$ and $1+q^{-2}$, and the triples with $u=1$ or $a=1$ contribute $O(\log X)$. The details are in Appendices A 1 and A 2. □

The two corners have the same constant because they have the same local factors (Table III in Appendix A 1).

B. Primes

To count the corners over the primes we use dyadic blocks. Fix $\delta \in (0, \frac{1}{10}]$. For $j \ge 1$ put

$$I_j = (2^j, 2^{j+1}], \qquad X_j = 2^{j(1/2-\delta)}.$$

For $p \in I_j$ we have $16X_j^2 \le 16\cdot 2^{j(1-2\delta)} \le p$ once $j \ge 2/\delta$. We write $|K_p(X)|$ for the size of a corner, with the convention $|K_p(X)| = 0$ when $p \not\equiv 3 \pmod 4$.

Proposition V.2. *For $K \in \{A,B,C\}$, as $j \to \infty$,*

$$\sum_{p\in I_j} \frac{|K_p(X_j)|}{p} = \frac{\kappa}{2}\Big(\frac{1}{2}-\delta\Big)^2 (\log 2)^2\, j + O_\delta\big(j^{1/2}\big).$$

Proof. For C the statement is the one for B, since $|C_p(X)| = |B_p(X)|$. Let $x = 2^{j+1}$ and $z = j^{1/2}$.

Corner A. A pair (u,a) with $u,a \ge 2$, $\gcd(u,a)=1$ and $ua \le X_j$ belongs to $A_p(X_j)$ exactly when $p \equiv -1 \pmod{4ua}$. Hence, for $y \in I_j$,

$$\sum_{2^j < p \le y} |A_p(X_j)| = \sum_{(u,a)} \big(\pi(y;4ua,-1) - \pi(2^j;4ua,-1)\big).$$

The moduli $4ua \le 4X_j$ are below $x^{1/2}(\log x)^{-B}$ for j large, and each modulus $4n$ arises from at most $\tau(n)$ pairs. The Bombieri–Vinogradov theorem [6], [8, Chapter 28] with these multiplicities (Appendix A 3) gives

$$\sum_{2^j < p \le y} |A_p(X_j)| = \big(\pi(y) - \pi(2^j)\big) S_A(X_j) + O\big(x(\log x)^{-3}\big)$$

uniformly in $y \in I_j$.

Corner B. Let $P_z = \prod_{q\le z} q$. A pair (u,α) with $u\alpha \le X_j$ belongs to $B_p(X_j)$ when $p \equiv -1 \pmod{4u\alpha}$ and $\gcd(u,a)=1$, where $a = (p+1)/(4u\alpha)$. For a squarefree $d \mid u$ the condition $d \mid a$ is $p \equiv -1 \pmod{4u\alpha d}$. Inclusion and exclusion over the primes $q \le z$ that divide u count the pairs with no common prime factor $q \le z$:

$$|B_p^{(z)}| = \sum_{u\alpha \le X_j} \sum_{d \mid (u, P_z)} \mu(d)\, \mathbf{1}\big[p \equiv -1 \pmod{4u\alpha d}\big].$$

These moduli are at most $4X_j e^{2z}$, still in the range of the Bombieri–Vinogradov theorem, and a modulus $4n$ arises at most $\tau_3(n)$ times. The main term is $(\pi(y)-\pi(2^j))S_B^{(z)}(X_j)$, where $S_B^{(z)}$ is S_B with $\varphi(u)/u$ replaced by $\prod_{q\mid u, q\le z}(1-1/q)$, and $S_B \le S_B^{(z)} \le S_B + O(\log^2 X_j/z)$ (Appendix A 6). Then $|B_p(X_j)| \le |B_p^{(z)}|$, and the difference counts pairs with $q \mid u$ and $q \mid a$ for some prime $q > z$. The Brun–Titchmarsh inequality [20] bounds their total over $p \in I_j$ by $O(\delta^{-1}\pi(I_j)\log^2 X_j/z)$ (Appendix A 6).

The weights $1/p$. Partial summation inside the block (Appendix A 12) turns the counts into

$$\sum_{p\in I_j} \frac{|K_p(X_j)|}{p} = S_K(X_j) \sum_{p\in I_j} \frac{1}{p} + O\Big(\frac{j}{\delta z}\Big) + O(1),$$

with $S_C = S_B$. By the prime number theorem with error term, $\sum_{p\in I_j} 1/p = 1/j + O(1/j^2)$. With $S_K(X_j) = \frac{\kappa}{2}(\frac{1}{2}-\delta)^2(\log 2)^2 j^2 + O(j)$ from Lemma V.1 and $z = j^{1/2}$, this is the statement. □

For $p \in I_j$ with $j \ge 2/\delta$ let $\omega_\delta^{\mathrm{cor}}(p)$ be the number of distinct classes $-\mathrm{c}(t)$ with t in one of the three corners at level X_j, and let $\omega_\delta^{\mathrm{cor}}(p) = 0$ for all other primes. For a function ω on the primes define

$$N[\omega](\lambda) = \sum_{p \le e^\lambda} \frac{\omega(p)}{p}. \tag{23}$$

[6] For a prime $q \mid u$, the primes $p \equiv -1 \pmod{4u\alpha}$ fall into q classes modulo $4u\alpha q$, all prime to $4u\alpha q$, and $q \mid a = (p+1)/(4u\alpha)$ holds on exactly one of them. By inclusion and exclusion over the prime factors of u, the condition $\gcd(u,a)=1$ has density $\prod_{q\mid u}(1-1/q) = \varphi(u)/u$, because $\varphi(4u\alpha q) = q\,\varphi(4u\alpha)$ when $q \mid u$.

Corollary V.3. *As $\lambda\to\infty$, $N[\omega_\delta^{\mathrm{cor}}](\lambda)=(a_\delta^{\mathrm{cor}}+o(1))\lambda^2$ with*

$$a_\delta^{\mathrm{cor}}=\frac{3\kappa}{4}\Big(\frac12-\delta\Big)^2.$$

In particular $a_\delta^{\mathrm{cor}}\to a_0=3\kappa/16=27/(32\pi^2)$ as $\delta\to0$.

Proof. By Propositions III.1 and III.2,

$$\sum_K|K_p(X_j)|-\nu(p)\le\omega_\delta^{\mathrm{cor}}(p)\le\sum_K|K_p(X_j)|.$$

Summing Proposition V.2 over the three corners and over $2/\delta\le j\le J$ gives

$$\frac{3\kappa}{2}\Big(\frac12-\delta\Big)^2(\log2)^2\sum_{j\le J}j+O_\delta(J^{3/2})$$
$$=a_\delta^{\mathrm{cor}}(J\log2)^2+O_\delta(J^{3/2}).$$

The coincidences contribute $O(1)$ by Corollary III.3, and a partial block contributes $O(J)$. With $J\log2=\lambda+O(1)$ the corollary follows. □

VI. THE STRIPS OF THE CENTRE

A. Maynard's theorem

For $0<\vartheta<\frac{1}{42}$ put

$$\theta(\vartheta)=\min\Big(\frac{1}{10}-\frac{7\vartheta}{5},\ \frac12-19\vartheta\Big).\tag{24}$$

Theorem VI.1 (Maynard [16, Corollary 1.2])**.** *Let $b\in\mathbb{Z}$, $0<\vartheta<\frac{1}{42}$ and $0<\eta<(1-42\vartheta)/4$, and let $Q_{\vartheta,\eta}(y)$ be the set of $q\le y^{1/2+\vartheta}$ that have a divisor d with $y^{2\vartheta+\eta}<d<y^{\theta(\vartheta)-\eta}$. Then for every $A>0$,*

$$\sum_{\substack{q\in Q_{\vartheta,\eta}(y)\\(q,b)=1}}\Big|\pi(y;q,b)-\frac{\pi(y)}{\varphi(q)}\Big|\ll_{b,\eta,A}\frac{y}{(\log y)^A}.$$

We use it with $b=-1$, so that every q is prime to b.[7] The theorem asks for a divisor in the modulus in a prescribed range. For the moduli $4ua$ of the centre, the next lemma finds one from the factorisations of u and a.

B. A divisor in a window

Definition VI.2. Let $1<A<B$ and $G=\min(B/A,A)$. For $n\ge1$ let $T(n)$ be the product of the prime powers $q^e\parallel n$ with $q<G$, and let $q(n)$ be the largest prime factor of n in $[G,A]$, or $q(n)=1$ if there is none. A pair (u,v) of positive integers *passes the rule* for the window (A,B) if

(R1) $u\in(A,B)$ or $v\in(A,B)$; or

(R2) uv has a prime factor in (A,B); or

(R3) $q(uv)\,T(uv)>A$.

Lemma VI.3. *If (u,v) passes the rule for (A,B), then uv has a divisor in (A,B).*

Proof. Under (R1) or (R2) this is clear. Suppose (R3) holds, and let $r_1,\dots,r_m$ be the prime factors of $T(uv)$, listed with multiplicity; each is below G. Put $d_0=q(uv)$ and $d_i=d_0r_1\cdots r_i$. Since $q(uv)\ge G$ when it is a prime, d_0 is prime to $T(uv)$, so every d_i divides uv. Now $d_0\le A<d_m$, so there is a least $i\ge1$ with $d_i>A$, and then

$$d_i=d_{i-1}r_i<A\,G\le A\cdot\frac{B}{A}=B.$$

So d_i is a divisor of uv in (A,B). □

The bound $r_i<G\le B/A$ on the steps of the chain is needed.[8] The condition $G\le A$ keeps the primes of $T(uv)$ apart from $q(uv)$.

C. The density of the pairs that fail

Let $0<\theta^-<\theta^+$ and put $\xi=\min(\theta^+-\theta^-,\theta^-)$. For a window $(x^{\theta^-},x^{\theta^+})$ the number G of Definition VI.2 is x^ξ. For $w\ge\xi$ let

$$\mathscr{A}_w=(\xi,w]\cup[\theta^+,\infty),$$

and let ϖ_w be the function on $\mathbb{R}$ that vanishes on $(-\infty,\xi]$ and satisfies

$$r\,\varpi_w(r)=\mathbf{1}_{\mathscr{A}_w}(r)+\int_{\mathscr{A}_w\cap(0,r)}\varpi_w(r-z)\,dz\qquad(r>\xi).\tag{25}$$

The right side involves ϖ_w only on $(-\infty,r-\xi)$, so (25) determines ϖ_w on $(\xi,2\xi]$, then on $(2\xi,3\xi]$, and so on. Let ρ be Dickman's function, with $\rho(v)=1$ for $0\le v\le1$ and $v\rho'(v)=-\rho(v-1)$ for $v>1$. For $s_1,s_2>\theta^-$ put $U(s_1,s_2)=0$ if s_1 or s_2 lies in (θ^-,θ^+), and otherwise

$$U(s_1,s_2)=\iint_\Delta\rho(t_1/\xi)\rho(t_2/\xi)\varpi_w(s_1-t_1)\varpi_w(s_2-t_2)\,dt_1dt_2,\tag{26}$$

where $\Delta=\{t_1,t_2\ge0,\ t_1+t_2\le\theta^-\}$ and $w=\max(\theta^--t_1-t_2,\xi)$. Then $0\le U\le1$ (Appendix A 8).

The function U is the density of the pairs that fail the rule. A number u of size x^{s_1} splits as $u=T(u)\,r$, where the x^ξ-smooth part $T(u)$ has size x^{t_1} with density $\rho(t_1/\xi)$, and the rough part

[7] The range $(y^{2\vartheta},y^{\theta(\vartheta)})$ for the divisor is not empty exactly when $\vartheta<\frac{1}{42}$, and its upper end is $y^{1/10-7\vartheta/5}$ exactly when $\vartheta\le\frac{1}{44}$. Its width is at least its lower end, $\theta(\vartheta)-2\vartheta\ge2\vartheta$, exactly when $\vartheta\le\frac{1}{54}$. These three values are where the integrand of (30) changes its shape (Figure 5).

[8] If $T(n)$ collected all the prime factors below A, the integer 4 and the window $(\frac52,\frac72)$ would pass (R3) with $T(4)=4>\frac52$, while none of the divisors 1, 2, 4 lies in the window. A brute-force check finds this failure 12,022,742 times among the integers up to $2\cdot10^6$ and 64 windows, and no failure of Lemma VI.3 (Section XII).

r has prime factors x^{z_i} with $z_i \ge \xi$, distributed with density $\prod_i dz_i/z_i$ by Mertens' theorem.[9] The pair fails (R1) and (R2) when $s_1, s_2 \notin (\theta^-, \theta^+)$ and no z_i lies in (θ^-, θ^+), and it fails (R3) when $t_1 + t_2 \le \theta^-$ and every z_i in $[\xi, \theta^-]$ is at most $\theta^- - t_1 - t_2$. These conditions say that all the z_i lie in $\mathscr{A}_w$.

Lemma VI.4. *Let $0 < \theta^- < \theta^+$, let U be the function (26), and let $R \subset (\theta^-, K)^2$, for some K, be a set whose boundary has Lebesgue measure zero. Let $h : \mathbb{N}^2 \to \mathbb{R}$ satisfy $\sum_{d,e} |h(d,e)|/(de) < \infty$, and put*

$$W(u,v) = \sum_{d \mid u,\, e \mid v} h(d,e), \qquad H = \sum_{d,e} \frac{h(d,e)}{de}.$$

Then, as $x \to \infty$,

$$\sum_{(u,v)} \frac{W(u,v)}{uv} = H(\log x)^2 \iint_R (1-U)\, ds_1\, ds_2 + o\big((\log x)^2\big),$$

where (u,v) runs over the pairs that pass the rule for $(x^{\theta^-}, x^{\theta^+})$ and satisfy $(\log u, \log v)/\log x \in R$.

The proof is in Appendix A 8.

D. The pieces

Fix $\delta \in (0, \frac{1}{200}]$ and a partition

$$\delta = \vartheta_0 < \vartheta_1 < \cdots < \vartheta_m = \frac{1}{42} - 2\delta, \qquad \vartheta_{i+1} - \vartheta_i \le \delta.$$

For $0 \le i < m$ put

$$\vartheta_i^+ = \vartheta_{i+1} + \delta, \qquad \theta_i^- = 2\vartheta_i^+ + \delta, \qquad \theta_i^+ = \theta(\vartheta_i^+) - \delta.$$

Since $\vartheta_i^+ \le \frac{1}{42} - \delta$, we have $\frac{1}{10} - \frac{17}{5}\vartheta_i^+ \ge \frac{2}{105}$ and $\frac{1}{2} - 21\vartheta_i^+ \ge 21\delta$, so $\theta_i^+ - \theta_i^- = \theta(\vartheta_i^+) - 2\vartheta_i^+ - 2\delta > 0$. As before, $x = 2^j$ for the block $I_j = (x, 2x]$.

Definition VI.5. For a prime $p \equiv 3 \pmod 4$ in I_j and $0 \le i < m$, let $\mathscr{Z}_{p,i}^A$ be the set of triples $t = (u, a, \alpha) \in \mathscr{T}(p)$ with

$$x^{1/2+\vartheta_i} < ua \le x^{1/2+\vartheta_{i+1}}, \qquad u, a > x^{\theta_i^-},$$

such that (u,a) passes the rule for the window $(x^{\theta_i^-}, x^{\theta_i^+})$. Let $\mathscr{Z}_{p,i}^B$ be defined in the same way with the pair (u,α) in place of (u,a), and let $\mathscr{Z}_{p,i}^C = \iota(\mathscr{Z}_{p,i}^B)$, defined with the pair (a,α).

[9] For a number counted with the weight $1/n$ the logarithmic sizes of its prime factors above x^ξ behave like the points of a Poisson process with intensity dz/z on $[\xi,\infty)$, and its x^ξ-smooth part is independent of them; this is the content of the theorems of Mertens and of Dickman and de Bruijn [26, Chapter III.5]. Equation (25) is the renewal equation for the density of $\sum_i z_i$ when every z_i is restricted to $\mathscr{A}_w$; with $\mathscr{A}_w$ replaced by (ξ,∞) it is Buchstab's equation.

Lemma VI.6. *Let j be large in terms of δ and $p \in I_j$. Then every piece $\mathscr{Z}_{p,i}^K$ lies in the centre $\mathscr{Z}_p$. In $\mathscr{Z}_{p,i}^A$ the largest coordinate is α, in $\mathscr{Z}_{p,i}^B$ it is a, and in $\mathscr{Z}_{p,i}^C$ it is u. The pieces $\mathscr{Z}_{p,i}^K$, for $K \in \{A,B,C\}$ and $0 \le i < m$, are pairwise disjoint.*

Proof. Let $t \in \mathscr{Z}_{p,i}^A$. Since $x < p \le 2x$ we have $x/4 < P < x$. Since $\theta_i^- = 2\vartheta_{i+1} + 3\delta$,

$$u = \frac{ua}{a} < x^{1/2+\vartheta_{i+1}-\theta_i^-} = x^{1/2-\vartheta_{i+1}-3\delta},$$

so $u^2 < x^{1-8\delta} < x/4 < P$ for j large, and in the same way $a^2 < P$. Next $\alpha = P/(ua) < Px^{-1/2}$, so $\alpha^2 < P \cdot P/x < P$. Hence $t \in \mathscr{Z}_p$. Moreover $\alpha \ge P/x^{1/2+\vartheta_{i+1}} > \frac{1}{4}x^{1/2-\vartheta_{i+1}}$, which exceeds u and a for j large. The pieces $\mathscr{Z}_{p,i}^B$ and $\mathscr{Z}_{p,i}^C$ are treated in the same way. Pieces with different K are disjoint because their largest coordinates differ, and pieces with the same K and different i are disjoint because the ranges of the relevant product are disjoint. □

Proposition VI.7. *For $K \in \{A,B,C\}$ and $0 \le i < m$, as $j \to \infty$,*

$$\sum_{p \in I_j} \frac{|\mathscr{Z}_{p,i}^K|}{p} = \kappa(\log 2)^2\, \mathscr{M}_i\, j + o(j),$$

where $\mathscr{M}_i = \iint_{R_i} (1 - U_i)\, ds_1\, ds_2$, with

$$R_i = \Big\{ (s_1,s_2) : \tfrac{1}{2} + \vartheta_i < s_1 + s_2 \le \tfrac{1}{2} + \vartheta_{i+1},\ s_1, s_2 > \theta_i^- \Big\}$$

and U_i the function (26) for the window (θ_i^-, θ_i^+).

Proof. For C the statement is the one for B, since ι is a bijection from $\mathscr{Z}_{p,i}^B$ onto $\mathscr{Z}_{p,i}^C$. Write $\eta = \delta/2$.

The moduli are in Maynard's range. Let (u,a) satisfy the conditions of Definition VI.5 and let $y \in [x, 2x]$. The modulus $q = 4ua$ satisfies $q \le 4x^{1/2+\vartheta_{i+1}} \le y^{1/2+\vartheta_i^+}$ for j large. By Lemma VI.3, ua has a divisor d with $x^{\theta_i^-} < d < x^{\theta_i^+}$, and for j large

$$y^{2\vartheta_i^+ + \eta} < x^{\theta_i^-}, \qquad x^{\theta_i^+} < y^{\theta(\vartheta_i^+) - \eta}.$$

Since $1 - 42\vartheta_i^+ \ge 42\delta$, we have $\eta < (1 - 42\vartheta_i^+)/4$. Hence $q \in Q_{\vartheta_i^+, \eta}(y)$.

The piece A. A coprime pair (u,a) as above gives a triple of $\mathscr{Z}_{p,i}^A$ exactly when $p \equiv -1 \pmod{4ua}$, and each modulus $4n$ arises from at most $\tau(n)$ pairs. Theorem VI.1 with these multiplicities, applied at the points $y_l = x(1 + l/j)$ for $0 \le l \le j$, and partial summation over the subintervals $(y_l, y_{l+1}]$ (Appendix A 4) give

$$\sum_{p \in I_j} \frac{|\mathscr{Z}_{p,i}^A|}{p} = \Big(\sum_{p \in I_j} \frac{1}{p} \Big) \sum_{(u,a)} \frac{1}{\varphi(4ua)} + O(1),$$

where (u,a) runs over the coprime pairs that pass the rule with $(\log u, \log a)/\log x \in R_i$. By (A1), $1/\varphi(4ua) = W_A(u,a)/(2ua)$ with $W_A(u,a) = \psi(ua)\mathbf{1}[\gcd(u,a) = 1]$, which has the form of Lemma VI.4 with $H = 2\kappa$ (Appendix A 1).

Lemma VI.4 gives $\sum_{(u,a)} 1/\varphi(4ua) = \kappa(j\log 2)^2\mathscr{M}_i + o(j^2)$, and $\sum_{p\in I_j} 1/p = 1/j + O(1/j^2)$.

The piece B. A pair (u,α) gives a triple of $\mathscr{Z}^B_{p,i}$ when $p \equiv -1 \pmod{4u\alpha}$ and $\gcd(u,a)=1$ with $a=(p+1)/(4u\alpha)$. As in Proposition V.2, inclusion and exclusion over $d \mid (u,P_z)$ with $z=j^{1/2}$ replaces this by the progressions modulo $4u\alpha d$. These moduli are at most $4x^{1/2+\vartheta_{i+1}}e^{2z} \le y^{1/2+\vartheta_i^+}$ and keep the divisor of $u\alpha$ found by Lemma VI.3, so they lie in $\mathcal{Q}_{\vartheta_i^+,\eta}(y)$, and a modulus $4n$ arises from at most $\tau_3(n)$ triples (u,α,d). The main term carries the weight $\psi(u\alpha)\prod_{q\mid u,\,q\le z}(1-1/q)$, which differs from $W_B(u,\alpha) = \psi(u\alpha)\varphi(u)/u$ by a total of $O(j^2/z)$ over the piece (Appendix A 6), and W_B has $H=2\kappa$. The pairs with a common prime factor $q>z$ of u and a contribute $O(j/z)+o(1)$ (Appendix A 7). The rest is as for A. □

VII. THE DEEP STRIPS

Below the depth $\frac{1}{42}$ a divisor of the modulus in a window lets Theorem VI.1 count the primes. Beyond it we use Maynard's second theorem [17], which reaches moduli up to $p^{3/5}$ when the weight attached to the moduli is a convolution of three factors of prescribed sizes. Whether such a splitting exists is decided by the sizes of the prime factors of the modulus, and this section counts the pairs for which it does.

A. Maynard's second theorem

Let $n\in\mathbb{R}$ and $\mu\ge 0$. A triple (d_1,d_2,d_3) of real numbers is *admissible for n with margin* μ if

$$\begin{aligned} d_1 < n-\mu, \quad & 2n+d_2+2d_3 < 1-\mu, \\ 2n+d_1+4d_2+3d_3 < 2-\mu, \quad & n+d_1+5d_2+2d_3 < 2-\mu. \end{aligned} \tag{27}$$

These are the four conditions of [17, Proposition 8.2] for a variable of size $N=y^n$ and factors of the moduli of sizes at most $Q_i = y^{d_i}$, with $\varepsilon=\mu/15$.[10] They become weaker when any d_i decreases.

Theorem VII.1 (Maynard [17]). *Let $0<\vartheta<\frac16$, $0<\varepsilon<\frac{1}{100}$ and $B,J\ge 1$, and put $\mathscr{N}_\vartheta = [\frac16-\vartheta-4\varepsilon, \frac13+4\varepsilon]$. Let (λ_q) be complex numbers supported on $q\le y^{1/2+\vartheta}$ with $|\lambda_q|\le\tau(q)^B$. Suppose that for every $n\in\mathscr{N}_\vartheta$ there are sequences $\gamma^{(j)}$, $\beta^{(j)}$, $\nu^{(j)}$, $1\le j\le J$, bounded by τ^B and supported on $[1,y^{d_1^{(j)}}]$, $[1,y^{d_2^{(j)}}]$ and $[1,y^{d_3^{(j)}}]$ for triples $d^{(j)}$ admissible for n with margin 15ε, such that*

$$\lambda - \sum_{j=1}^{J} \gamma^{(j)} * \beta^{(j)} * \nu^{(j)}$$

is bounded by τ^B and supported on the moduli divisible by p'^2 for a prime $p'>y^\eta$, where $\eta>0$ is fixed. Then, for every $A>0$,

$$\sum_q \lambda_q\Big(\pi(y;q,-1) - \frac{\pi(y)}{\varphi(q)}\Big) \ll_{A,B,J,\varepsilon,\eta} \frac{y}{(\log y)^A}.$$

Maynard states this for weights that are triply well factorable of level $y^{3/5-\varepsilon}$, with the range $[\varepsilon,\frac25]$ for n [17, Theorem 1.1], and his Section 9 applies the same reduction to the linear sieve. The proof uses his Propositions 5.2 and 8.2 and the identity of Heath-Brown; Appendix A 5 gives it for the range $\mathscr{N}_\vartheta$, whose lower end comes from the small variables of that identity.[11]

B. Factorable profiles

Fix $G\in(0,\frac{1}{10})$. The G-profile of an integer n is the pair (t,Z) of Appendix A 8: t is the logarithmic size of the part of n made of the primes below x^G, and Z is the multiset of the logarithmic sizes of the prime factors of n from x^G on. The combined profile of a pair (u,v) is $(t_1+t_2, Z_1\cup Z_2)$, which is the profile of uv when $\gcd(u,v)=1$.

Definition VII.2. Let (t,Z) be a G-profile of size $s=t+\sum Z$ and let $\mu\ge 0$. An *assignment* for (t,Z) is a partition $Z = Z_1\sqcup Z_2\sqcup Z_3$, with sums b_1,b_2,b_3, together with an index $L\in\{1,2,3\}$, a set $F\subset\{1,2,3\}\setminus\{L\}$ and numbers $x_i\ge b_i$ for $i\in F$ such that

$$\sum_{i\in F}(x_i-b_i) + (|F|-1)^+G < t-\mu. \tag{28}$$

It *serves n* with margin μ if every triple (d_1,d_2,d_3) with $d_i\in[x_i,x_i+G]$ for $i\in F$, $d_i=b_i$ for $i\notin F\cup\{L\}$ and $d_L = s-\sum_{i\ne L}d_i$ is admissible for n with margin μ. The profile is *factorable with margin* μ if no element of Z lies within μ of G and every $n\in[\frac23-s-\mu,\frac13+\mu]$ is served by some assignment with margin μ. A pair of integers, or of profiles, is *factorable* if for some $G\in(0,\frac{1}{10})$ its combined G-profile is factorable with some margin $\mu>0$.

An assignment describes a splitting of the modulus $q=4uv$ into three factors. The primes from x^G on are placed according to Z_1,Z_2,Z_3, and the primes below x^G, whose total size is t, are added one at a time to the groups $i\in F$ until each reaches

[10] In [17] they read $Q_1<N/y^\varepsilon$, $N^2Q_2Q_3^2<y^{1-8\varepsilon}$, $N^2Q_1Q_2^4Q_3^3<y^{2-15\varepsilon}$ and $NQ_1Q_2^5Q_3^2<y^{2-15\varepsilon}$. Each of them becomes weaker when one of the Q_i decreases, so a triple that bounds the sizes of the factors from above may be used in place of their exact sizes.

[11] The lower end $\frac16-\vartheta$ is $\frac23$ minus the level $\frac12+\vartheta$. A product of variables of the identity that are all below $y^{1/6-\vartheta}$ stays below it, because the range $\mathscr{N}_\vartheta$ is longer than its lower end, and such a product is handled together with two large variables by Maynard's estimate for the divisor function, which needs moduli at most $y^{2/3}$ divided by the product.

the size x_i, which it overshoots by less than G; the condition (28) leaves enough of them for the second group, and the rest go to the group L. The definition asks for the conditions of Theorem VII.1 for every size the factors can take in this process (Figure 3).[12]

Lemma VII.3. *Let (t,Z) be a G-profile of size s with no element equal to G, and suppose that every $n \in [\frac{2}{3}-s, \frac{1}{3}]$ is served by some assignment with margin 0. Then (t,Z) is factorable with some margin $\mu > 0$.*

Proof. Let $n_0 \in [\frac{2}{3}-s, \frac{1}{3}]$ be served by an assignment with margin 0. The inequality (28) is strict, and the triples in the definition form a compact set on which the strict inequalities (27) hold, so the assignment serves n_0 with some margin $\mu_0 > 0$. The left sides of (27) change by at most $2|n-n_0|$ when n_0 is replaced by n, so the assignment serves every n with $|n-n_0| < \mu_0/4$ with margin $\mu_0/2$. Finitely many of these intervals cover $[\frac{2}{3}-s, \frac{1}{3}]$, and their union contains a closed interval that is larger by some $\mu_1 > 0$ at each end. The least of μ_1, the margins $\mu_0/2$ of the finitely many assignments and the distance from Z to G is a margin for (t,Z). □

Let D be the set of (s_1,s_2) with $s_1+s_2 \ge \frac{1}{2}+\frac{1}{42}$, $2s_1+s_2<1$ and $s_1+2s_2<1$; in the coordinates (s_1,ϑ) with $\vartheta = s_1+s_2-\frac{1}{2}$ it is the part of the strips $2\vartheta < s_1 < \frac{1}{2}-\vartheta$ with $\frac{1}{42} \le \vartheta < \frac{1}{6}$. Put

$$\mathscr{K} = \lambda_1 \otimes \lambda_1 \big(\{ (\Pi(u), \Pi(v)) \text{ factorable, sizes in } D \} \big), \quad (29)$$

the measure, for the product of the measures λ_1 of Appendix A 8 on the ξ-profiles of u and v with ξ small, of the factorable pairs whose sizes lie in D. The set of factorable pairs is the union over the rational G of the sets defined with a fixed G, since a profile that is factorable for G with margin μ is factorable for every $G' \in [G, G+\mu/10]$ with margin $\mu/2$,[13] and it does not depend on ξ once $\xi \le G$. Since the image of λ_1 under the size is Lebesgue measure, $0 \le \mathscr{K} \le \operatorname{area}(D) = \frac{3}{98}$.[14]

The deep strips cannot be counted in this way beyond the depth $\frac{1}{10}$.

Lemma VII.4. *Let $s = \frac{1}{2}+\vartheta$ with $\vartheta \ge \frac{1}{10}$. For no n is there a triple (d_1,d_2,d_3) with $d_1+d_2+d_3 = s$ that is admissible for n with margin 0. Hence no profile of size s is factorable, and*

$$\mathscr{K} = \lambda_1 \otimes \lambda_1 \big(\{ (\Pi(u), \Pi(v)) \textit{ factorable}, \textit{ sizes in } D,\ \vartheta < \tfrac{1}{10} \} \big) \le \frac{88}{3675}.$$

Proof. Put $d_3 = s-d_1-d_2$ in the second and third conditions of (27). With margin 0 they read

$$d_2 > 2s-1+2(n-d_1), \qquad d_2 < 2-3s-2(n-d_1),$$

and the first condition gives $n-d_1 > 0$. Hence $2s-1 < d_2 < 2-3s$, which needs $5s<3$, that is $\vartheta < \frac{1}{10}$. A factorable profile of size s has an assignment that serves $n = \frac{1}{3}$, and every triple of that assignment has sum s and is admissible. The bound is the area $\int_{1/42}^{1/10} (\frac{1}{2}-3\vartheta)\,d\vartheta = \frac{88}{3675}$ of the part of D below $\vartheta = \frac{1}{10}$.[15] □

C. The deep pieces

Fix $\delta \in (0, \frac{1}{200}]$ as in Section VI, a finite set $\mathscr{G} \subset (0, \frac{1}{10})$, a margin $\mu \in (0, \frac{1}{100})$ and $\varepsilon_1 \in (0, \delta^2)$ with $\varepsilon_1 < \mu^2$, and put $\varepsilon_0 = \varepsilon_1^2$ and $h = \varepsilon_0^2$. For a prime $q \ge x^{\varepsilon_0}$ let its *box* be $k(q) = \lceil \log q/(h \log x) \rceil$ and its *nominal size* $k(q)h$, so that $\log q/\log x \in ((k-1)h, kh]$. Every $n \ge 1$ factors as $n = D(n)r(n)$, where $D(n)$, the *dust* of n, is the part made of the primes below x^{ε_0}. The *box datum* $\beta(n)$ is the multiset of the boxes of the prime factors of $r(n)$, with multiplicity, and $\sigma(n) = \sum_{k \in \beta(n)} kh$.

Definition VII.5. A pair of box data (β_1,β_2) with nominal sizes σ_1,σ_2 is *good* if $2\sigma_1+\sigma_2 < 1-4\varepsilon_1$, $\sigma_1+2\sigma_2 < 1-4\varepsilon_1$, and for some $G \in \mathscr{G}$ the nominal profile

$$\Big(2\varepsilon_1 + \sum_{\substack{k \in \beta_1 \cup \beta_2 \\ kh < G}} kh,\ \{ kh : k \in \beta_1 \cup \beta_2,\ kh \ge G \} \Big)$$

is factorable with margin μ for this G. Let $\frac{1}{42} = \vartheta_0' < \vartheta_1' < \cdots < \vartheta_{m'}' = \frac{1}{6}$ be a partition with $\vartheta_{i+1}' - \vartheta_i' \le \delta$. For a prime $p \equiv 3 \pmod 4$ in I_j and $0 \le i < m'$, let $\mathscr{Y}_{p,i}^A$ be the set of triples $t = (u,a,\alpha) \in \mathscr{T}(p)$ such that $D(u), D(a) \le x^{\varepsilon_1}$, $r(u)r(a)$ is squarefree, $(\beta(u),\beta(a))$ is good, and

$$\tfrac{1}{2} + \vartheta_i' < \sigma(u) + \sigma(a) \le \tfrac{1}{2} + \vartheta_{i+1}'.$$

Let $\mathscr{Y}_{p,i}^B$ be defined in the same way with the pair (u,α) in place of (u,a), and $\mathscr{Y}_{p,i}^C = \iota(\mathscr{Y}_{p,i}^B)$, defined with (a,α).

Lemma VII.6. *Let j be large in terms of δ, $\mathscr{G}$, μ and ε_1, and let $p \in I_j$. Every piece $\mathscr{Y}_{p,i}^K$ lies in the centre $\mathscr{Z}_p$, with largest coordinate α, a and u for $K = A,B,C$. The pieces $\mathscr{Y}_{p,i}^K$ are pairwise disjoint and disjoint from the pieces $\mathscr{Z}_{p,i}^K$ of Definition VI.5.*

Proof. Let $t \in \mathscr{Y}_{p,i}^A$ and write $s_1 = \log u/\log x$, $s_2 = \log a/\log x$. A number n with $D(n) \le x^{\varepsilon_1}$ has at most $1/\varepsilon_0$ prime factors

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[12] The targets x_i are fixed numbers, so the size of a filled group is known only up to G, and the size of the group L is what remains. The four conditions (27) are affine, so it is enough to check them at the vertices of the box of possible sizes, which is how the computation of Appendix A 11 proceeds.

[13] The elements of Z lie above $G+\mu$, so t and Z do not change; the budget (28) and the sizes d_i change by at most $G'-G$, and each of the four conditions (27) by at most $5(G'-G)$.

[14] In the coordinates (s_1,ϑ) the area of D is $\int_{1/42}^{1/6} (\frac{1}{2}-3\vartheta)\,d\vartheta = \frac{1}{2}(\frac{1}{2}-\frac{1}{14})^2 \cdot \frac{1}{3} = \frac{3}{98}$, the triangle with vertices $(\frac{1}{21},\frac{1}{42})$, $(\frac{10}{21},\frac{1}{42})$ and $(\frac{1}{3},\frac{1}{6})$.

[15] In terms of the sizes: the second condition asks the modulus to have a factor q_2 of size above $p^{2\vartheta}$, the third asks q_2 to be below $p^{1/2-3\vartheta}$, and these two ranges meet only for $\vartheta < \frac{1}{10}$, whatever the size of the long variable.

from x^{ε_0} on, each of them below its nominal size and above it minus h, so

$$\sigma(n)-\varepsilon_0 \le \frac{\log n}{\log x} \le \sigma(n)+\varepsilon_1.$$

Hence $2s_1+s_2<1-\varepsilon_1$ and $s_1+2s_2<1-\varepsilon_1$. Since $\log\alpha/\log x = 1-s_1-s_2+O(1/j)$, this gives $u,a<\alpha$ for j large, and then $u^2<u\alpha\le P$ and $a^2<P$; also $s_1+s_2>\frac12+\frac{1}{42}-2\varepsilon_0$, so $\alpha^2<P$. Thus $t\in\mathscr{Z}_p$ with largest coordinate α, and the pieces B and C are treated in the same way. Pieces with different K have different largest coordinates; pieces with the same K and different i have disjoint ranges of $\sigma(u)+\sigma(a)$. Finally $\log(ua)/\log x>\frac12+\frac{1}{42}-2\varepsilon_0$ on the new pieces, while $ua\le x^{1/2+1/42-2\delta}$ on the pieces of Definition VI.5, and $2\varepsilon_0<2\delta$. □

Proposition VII.7. *For $K\in\{A,B,C\}$ and $0\le i<m'$, as $j\to\infty$,*

$$\sum_{p\in I_j}\frac{|\mathscr{Y}^K_{p,i}|}{p} = \kappa(\log 2)^2\,\mathscr{K}_i\,j+o(j),$$

where $\mathscr{K}_i=\lambda_1\otimes\lambda_1(E_i)$ and E_i is the set of pairs of ε_0-profiles with dust at most ε_1 whose box data form a good pair with $\frac12+\vartheta'_i<\sigma_1+\sigma_2\le\frac12+\vartheta'_{i+1}$.

Proof. For C the statement is the one for B. The set E_i is a finite union of sets of profile pairs defined by the numbers of prime factors in each box and by the bounds on the dust, so it is bounded and its boundary is $\lambda_1\otimes\lambda_1$-null.

The piece A. Let $\mathscr{B}$ be the finite set of good pairs (β_1,β_2) in the range of the piece, and for $\beta\in\mathscr{B}$ let λ^β_q be the number of pairs (u,a) with $4ua=q$, $(\beta(u),\beta(a))=\beta$, $D(u),D(a)\le x^{\varepsilon_1}$, $\gcd(D(u),D(a))=1$ and $r(u)r(a)$ squarefree. These are the coprime pairs with the conditions of Definition VII.5, so

$$|\mathscr{Y}^A_{p,i}| = \sum_{\beta\in\mathscr{B}}\sum_{q\mid p+1}\lambda^\beta_q.$$

Put $\sigma_\beta=\sigma_1+\sigma_2+2\varepsilon_1$. Since $\log n/\log x\le\sigma(n)+\varepsilon_1$, every modulus satisfies $q\le 4x^{\sigma_\beta}$. By Lemma A.2, for every $n\in[\frac23-\sigma_\beta-\mu,\frac13+\mu]$ the weight λ^β is a fixed multiple of a convolution of three sequences bounded by τ^2, whose supports have sizes admissible for n with margin μ, up to a sequence bounded by τ^6 and supported on the moduli divisible by the square of a prime above x^{ε_0}. We apply Theorem VII.1 at every $y\in[x,2x]$ with $\vartheta=\sigma_\beta-\frac12+\varepsilon_1/10$, $\varepsilon=\mu/40$ and $\eta=\varepsilon_0/2$: the moduli are below $y^{1/2+\vartheta}$, $0<\vartheta<\frac16$ because $\frac12+\frac{1}{42}<\sigma_\beta<\frac23-\frac23\varepsilon_1$, the range $\mathscr{N}_\vartheta$ lies in $[\frac23-\sigma_\beta-\mu,\frac13+\mu]$, and replacing x by y changes the sizes by $o(1)$, which the margin $\mu-15\varepsilon>0$ absorbs. Partial summation over the points $y_l=x(1+l/j)$, as in Appendix A 4, now gives

$$\sum_{p\in I_j}\frac{|\mathscr{Y}^A_{p,i}|}{p} = \Big(\sum_{p\in I_j}\frac1p\Big)\sum_{(u,a)}\frac{1}{\varphi(4ua)}+o(1),$$

where (u,a) runs over the pairs counted by the weights λ^β, $\beta\in\mathscr{B}$. These are the coprime pairs whose profile pair lies in E_i, except for the pairs in which $r(u)$ or $r(a)$ has a repeated prime factor, whose total weight is $O(x^{-\varepsilon_0}j^2)$. With W_A as in the proof of Proposition VI.7, Lemma A.4 gives $\sum_{(u,a)}1/\varphi(4ua)=\kappa(j\log 2)^2\mathscr{K}_i+o(j^2)$.

The piece B. Here λ^β counts the pairs (u,α) with the conditions of Definition VII.5, with no coprimality between $D(u)$ and $D(\alpha)$, and such a pair gives a triple of $\mathscr{Y}^B_{p,i}$ when $p\equiv-1 \pmod{4u\alpha}$ and $\gcd(u,a)=1$ with $a=(p+1)/(4u\alpha)$. As in Proposition VI.7, inclusion and exclusion over $d\mid(u,P_z)$ with $z=j^{1/2}$ replaces this by the progressions modulo $4u\alpha d$ with the weights $\mu(d)$. The prime factors of d lie below $z<x^{\varepsilon_0}$, so d divides the dust of u, and the weight on the moduli $4u\alpha d$ has the form of Lemma A.2 with a weight on the dust bounded by τ_3; the sizes change by $o(1)$. The pairs with a common prime factor of u and a above z contribute $O(j/z)+o(1)$ (Appendix A 7, whose bounds hold for $u\alpha\le x^{2/3}$), and the main term carries the weight of Proposition VI.7. The rest is as for A. □

Lemma VII.8. *For every $\eta>0$ the parameters $\mathscr{G}$, μ and ε_1 can be chosen, in this order, so that $\sum_i\mathscr{K}_i\ge\mathscr{K}-\eta$ for every partition (ϑ'_i).*

Proof. The sets E_i are disjoint and their union E does not depend on the partition. For a finite set $\mathscr{G}$ and $\mu>0$ let $A(\mathscr{G},\mu)$ be the set of pairs of profiles with sizes in D whose combined G-profile is factorable with margin 2μ for some $G\in\mathscr{G}$. These sets increase as $\mathscr{G}$ grows and μ decreases, and by the remark after (29) their union over the finite sets of rationals and over $\mu>0$ is the set of factorable pairs, so we may fix $\mathscr{G}$ and μ with $\lambda_1\otimes\lambda_1(A(\mathscr{G},\mu))>\mathscr{K}-\eta/2$. A pair in $A(\mathscr{G},\mu)$ lies in E if its dust is at most ε_1, its sizes lie at distance more than $5\varepsilon_1$ from the boundary of D, and none of its prime factors has size in $[G-h,G)$ for a $G\in\mathscr{G}$ (Lemma A.3, with ε_1 small in terms of μ). As $\varepsilon_1\to0$ the measure of the pairs excluded by these three conditions tends to 0: the dust exceeds ε_1 with probability $O(\rho(\varepsilon_1/\varepsilon_0))=O(\rho(1/\varepsilon_1))$, a prime factor falls in $[G-h,G)$ with probability $O(h/G)$ for each of the finitely many G, and the strip of width $5\varepsilon_1$ along the boundary of D has area $O(\varepsilon_1)$. Hence $\lambda_1\otimes\lambda_1(E)>\mathscr{K}-\eta$ for ε_1 small. □

D. The density of all the classes

For $p\in I_j$ with j large in terms of $\delta,\mathscr{G},\mu$ and ε_1, let $\omega_\delta(p)$ be the number of distinct classes $-\mathrm{c}(t)$ with t in one of the three corners at level X_j, in one of the pieces $\mathscr{Z}^K_{p,i}$ or in one of the pieces $\mathscr{Y}^K_{p,i}$, and let $\omega_\delta(p)=0$ for all other primes. The subscript records δ only; the other parameters are fixed with it.

Corollary VII.9. *As $\lambda\to\infty$, $N[\omega_\delta](\lambda)=(a_\delta+o(1))\lambda^2$ with*

$$a_\delta=\frac{3\kappa}{4}\Big(\frac12-\delta\Big)^2+\frac{3\kappa}{2}\sum_{i<m}\mathscr{M}_i+\frac{3\kappa}{2}\sum_{i<m'}\mathscr{K}_i.$$

(a) one assignment: prime factors above x^G (dark), below x^G (light)

(b) the ranges of n served by six assignments

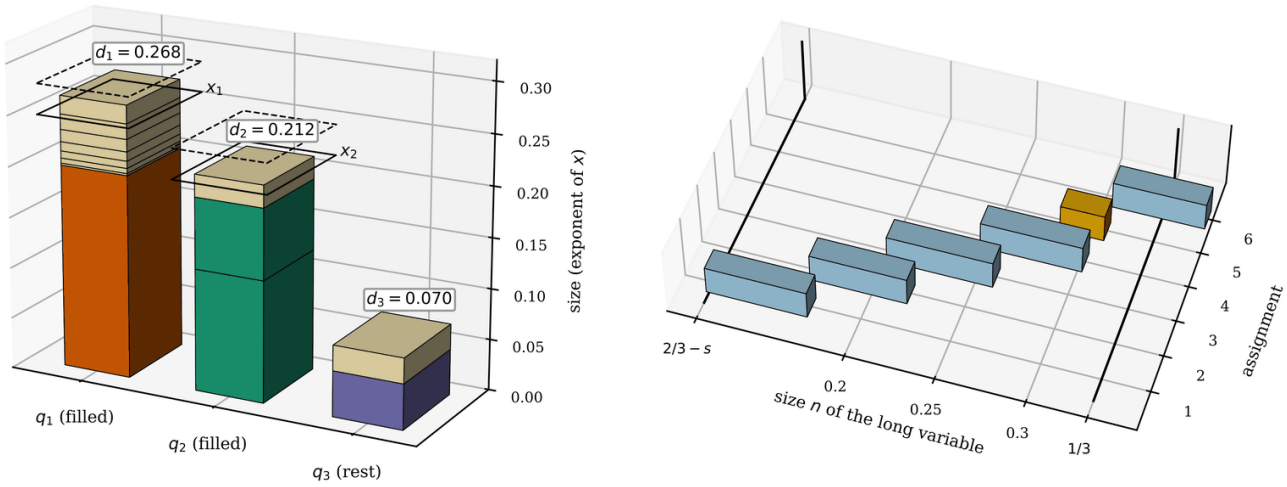


FIG. 3. Splitting a modulus $q = 4ua$ of size x^s, $s = 0.55$, into three factors $q_1q_2q_3$ for Theorem VII.1, with $G = 0.03$. (a) One assignment: the prime factors above x^G (dark) are placed in the groups Z_1, Z_2, Z_3; the primes below x^G (light) fill the groups 1 and 2 up to the targets x_1, x_2 (solid planes), which they overshoot by less than G (dashed planes), and the rest go to the group 3. (b) Each assignment serves an interval of sizes n of the long variable: there the sizes (d_1, d_2, d_3) satisfy the four inequalities (27) however the fill ends. Six assignments, the one of (a) among them, cover $[\frac{2}{3} - s, \frac{1}{3}]$, so this profile is factorable.

Proof. The corners at level $X_j \le \sqrt{p}/4$ are pairwise disjoint (Lemma II.5), the pieces $\mathscr{Z}^K_{p,i}$ and $\mathscr{Y}^K_{p,i}$ lie in the centre and are pairwise disjoint (Lemmas VI.6 and VII.6), and the centre is disjoint from the corners (Lemma IV.2). The classes are injective on each corner and on the centre, and the only coincidences between different regions are those counted by $\nu(p)$ and $\nu'(p)$ (Propositions III.1, IV.3 and IV.4). Hence

$$\Sigma_p - \nu(p) - \nu'(p) \le \omega_\delta(p) \le \Sigma_p,$$

where Σ_p is the sum of $|K_p(X_j)|$ over the three corners and of the sizes of all the pieces. Propositions V.2, VI.7 and VII.7, summed over the blocks $j \le J$, give

$$\sum_{p \le 2^{J+1}} \frac{\Sigma_p}{p} = a_\delta\,(J \log 2)^2 + o(J^2),$$

and the coincidences contribute $O(1)$ by Corollaries III.3 and IV.5. □

Lemma VII.10. *For $0 < \vartheta < \frac{1}{42}$ let U_ϑ be the function (26) for the window $(2\vartheta, \theta(\vartheta))$, and put*

$$\ell(\vartheta) = \int_{2\vartheta}^{1/2-\vartheta} U_\vartheta(s, \tfrac{1}{2} + \vartheta - s)\,ds, \quad \mathscr{J} = \int_0^{1/42} \ell(\vartheta)\,d\vartheta. \tag{30}$$

Then $\sum_i \mathscr{M}_i \to \frac{13}{1176} - \mathscr{J}$ as $\delta \to 0$, uniformly over the partitions of mesh at most δ. Consequently, for every $\eta > 0$ the parameters δ, $\mathscr{G}$, μ and ε_1 can be chosen so that $a_\delta > a_1 - \eta$, where

$$a_1 = \frac{\kappa}{4}\Big(\frac{40}{49} - 6\mathscr{J} + 6\mathscr{K}\Big).$$

Proof. The union of the regions R_i is the set of (s_1, s_2) with $\vartheta = s_1 + s_2 - \frac{1}{2} \in (\delta, \frac{1}{42} - 2\delta]$ and $s_1, s_2 > \theta^-_{i(\vartheta)}$, where $\theta_i^- = 2\vartheta + O(\delta)$. In the coordinates (s_1, ϑ) its area tends to

$$\int_0^{1/42} \Big(\frac{1}{2} - 3\vartheta\Big)\,d\vartheta = \frac{1}{84} - \frac{3}{3528} = \frac{13}{1176},$$

and $\sum_i \iint_{R_i} U_i$ tends to $\mathscr{J}$ by dominated convergence, because the windows (θ_i^-, θ_i^+) converge to $(2\vartheta, \theta(\vartheta))$ (Appendix A 8). Choose δ with $\frac{3\kappa}{4}(\frac{1}{2} - \delta)^2 + \frac{3\kappa}{2}\sum_i \mathscr{M}_i > \frac{3\kappa}{16} + \frac{3\kappa}{2}(\frac{13}{1176} - \mathscr{J}) - \frac{\eta}{2}$, and then $\mathscr{G}$, μ and $\varepsilon_1 < \delta^2$ by Lemma VII.8 with $\frac{3\kappa}{2}\sum_i \mathscr{K}_i > \frac{3\kappa}{2}\mathscr{K} - \frac{\eta}{2}$. Finally $\frac{3\kappa}{16} + \frac{3\kappa}{2}(\frac{13}{1176} - \mathscr{J}) + \frac{3\kappa}{2}\mathscr{K} = \frac{\kappa}{4}(\frac{40}{49} - 6\mathscr{J} + 6\mathscr{K})$. □

VIII. THE LARGE SIEVE AND THE TILT

Let $\omega(p)$ be integers with $0 \le \omega(p) < p$, and for each prime p let Ω_p be a set of $\omega(p)$ residue classes modulo p. Put $g(p) = \omega(p)/(p - \omega(p))$ and

$$L(Q) = \sum_{q \le Q} \mu^2(q) \prod_{p \mid q} g(p). \tag{31}$$

Lemma VIII.1 (Large sieve [19, 20])**.** *For all $N, Q \ge 1$, the number of $n \le N$ with $n \bmod p \notin \Omega_p$ for every prime $p \le Q$ is at most $(N + Q^2)/L(Q)$.*

The size of $L(Q)$ is determined by the density of the classes. We assume $\omega(p) \le C_\eta p^\eta$ for every $\eta > 0$; this holds in our applications because $\omega(p) \le |\mathscr{T}(p)| \le \tau_3(P)$. Let $N(\lambda) = N[\omega](\lambda)$ as in (23).

Proposition VIII.2. *Let $a>0$ and $T=\log Q$.*

(i) For every $\sigma>0$,

$$L(Q)\le Q^{\sigma}\exp F(\sigma),\qquad F(\sigma)=\sum_{p}g(p)p^{-\sigma}. \tag{32}$$

If $N(\lambda)\le(a+o(1))\lambda^2$, then $F(\sigma)\le(2a+o(1))\sigma^{-2}$ as $\sigma\to0^{+}$, and

$$\log L(Q)\le\frac{3}{2}(4a)^{1/3}T^{2/3}(1+o(1)).$$

(ii) If $N(\lambda)=(a+o(1))\lambda^2$, then

$$\log L(Q)\ge\frac{3}{2}(4a)^{1/3}T^{2/3}(1-o(1)).$$

Proof. (i) Since $(Q/q)^{\sigma}\ge1$ for $q\le Q$,

$$L(Q)\le Q^{\sigma}\sum_{q}\mu^2(q)\frac{\prod_{p\mid q}g(p)}{q^{\sigma}}=Q^{\sigma}\prod_{p}\bigl(1+g(p)p^{-\sigma}\bigr),$$

and $1+y\le e^{y}$ gives (32); this is Rankin's device [24]. The difference $g(p)-\omega(p)/p=\omega(p)^2/(p(p-\omega(p)))$ is summable, and partial summation gives $\sum_p(\omega(p)/p)p^{-\sigma}=\sigma\int_0^{\infty}e^{-\sigma\lambda}N(\lambda)\,d\lambda\le(2a+o(1))\sigma^{-2}$ (Appendix A 13). The choice $\sigma^3T=4a$ minimises $\sigma T+2a/\sigma^2$, with value $\frac{3}{2}\sigma T=\frac{3}{2}(4a)^{1/3}T^{2/3}$.

(ii) Fix $\sigma>0$ and let X_p, for $p\le Q$, be independent random variables with $X_p=1$ with probability $\pi_p=y_p/(1+y_p)$, where $y_p=g(p)p^{-\sigma}$, and $X_p=0$ otherwise. The random squarefree number $q=\prod_p p^{X_p}$ takes the value $q\mid\prod_{p\le Q}p$ with probability $\prod_{p\mid q}y_p/\prod_{p\le Q}(1+y_p)$. Let μ_σ and V_σ be the mean and variance of $\log q=\sum_p X_p\log p$. Suppose $\mu_\sigma+\sqrt{2V_\sigma}=T$. By Chebyshev's inequality, $|\log q-\mu_\sigma|<\sqrt{2V_\sigma}$ with probability at least $\frac12$, and then $q\le Q$ and $q^{\sigma}\ge e^{\sigma(T-2\sqrt{2V_\sigma})}$. Hence

$$\begin{aligned}L(Q)&\ge\sum_{\substack{q\le Q\\ |\log q-\mu_\sigma|<\sqrt{2V_\sigma}}}\mu^2(q)\frac{\prod_{p\mid q}g(p)}{q^{\sigma}}\,q^{\sigma}\\&\ge\frac12\,e^{\sigma T-2\sigma\sqrt{2V_\sigma}}\prod_{p\le Q}(1+y_p).\end{aligned} \tag{33}$$

In Appendix A 14 we show that such a σ exists, that $\sigma T\to\infty$ and $\sigma\asymp T^{-1/3}$, that $V_\sigma\ll\sigma^{-4}$, and that $\sum_{p\le Q}\log(1+y_p)\ge(2a-o(1))\sigma^{-2}$. Then (33) gives

$$\log L(Q)\ge\sigma T+\frac{2a-o(1)}{\sigma^2}-O(T^{1/3}),$$

and $\sigma T+2a'/\sigma^2\ge\frac32(4a')^{1/3}T^{2/3}$ for every $\sigma>0$ and $a'>0$. □

The lower bound (33) is an exponential tilt: the weight $q^{-\sigma}$ moves the distribution of $\log q$ so that its mean sits at $\log Q$, and the sum of the large sieve is then comparable to the whole Euler product.[16]

[16] The same picture explains the shape of the bound. The Euler product grows like $\exp(2a/\sigma^2)$, the tilt costs $Q^{-\sigma}$, and the balance $\sigma\asymp(\log Q)^{-1/3}$ gives the exponent $(\log Q)^{2/3}$. With $Q=\sqrt{N}$ the factor $(\frac12)^{2/3}$ combines with $(4a)^{1/3}$ to give $a^{1/3}$, which is why the final constant is $\frac32a^{1/3}$.

IX. PROOFS OF THE THEOREMS

Proof of Theorem A. Fix $\delta\in(0,\frac{1}{200}]$, the partitions of mesh at most δ of Sections VI and VII, and the parameters $\mathscr{G}$, μ and ε_1 of Definition VII.5. For $p\in I_j$ with j large, take Ω_p to be the set of the $\omega_\delta(p)$ classes $-\mathrm{c}(t)$ with t in the corners at level X_j, in the pieces $\mathscr{Z}^K_{p,i}$ or in the pieces $\mathscr{Y}^K_{p,i}$, and take Ω_p empty otherwise. By Lemma II.2, every $n\in\mathscr{E}(N)$ avoids these classes for all p. Lemma VIII.1 with $Q=\sqrt{N}$ gives

$$|\mathscr{E}(N)|\le\frac{2N}{L_\delta(\sqrt{N})}.$$

By Corollary VII.9 and Proposition VIII.2(ii) with $T=\frac12\log N$,

$$\log L_\delta(\sqrt{N})\ge\frac32(4a_\delta)^{1/3}\Bigl(\frac{\log N}{2}\Bigr)^{2/3}(1-o(1)).$$

Since $(4a_\delta)^{1/3}2^{-2/3}=a_\delta^{1/3}$, this is $\frac32a_\delta^{1/3}(\log N)^{2/3}(1-o(1))$. Since $\kappa/4=9/(8\pi^2)$, the number $\frac32a_1^{1/3}$ of Lemma VII.10 is c_1. Given $\varepsilon>0$, Lemma VII.10 lets us choose the parameters with $\frac32a_\delta^{1/3}>c_1-\varepsilon/2$; then (3) holds for all large N. The bound $\mathscr{J}<0.000335$ is Proposition X.2, the bounds $\mathscr{K}>0.00859$ and $c_1>0.6931$ are Proposition X.3, the bound (5) is Proposition X.1, and $\mathscr{J}\ge0$ because $U\ge0$. □

Theorem B needs only an upper bound for the density of the classes $\Omega_p^{\sharp}$, which come from triples with one product at most $\sqrt{p}$. Let $\omega^{\sharp}(p)=|\Omega_p^{\sharp}|$ and $N^{\sharp}(\lambda)=N[\omega^{\sharp}](\lambda)$.

Proposition IX.1. $N^{\sharp}(\lambda)\le(a_0+o(1))\lambda^2$.

Proof. We have $\omega^{\sharp}(p)\le|A_p^{\sharp}|+|B_p(\sqrt{p})|+|C_p(\sqrt{p})|$, where $A_p^{\sharp}$ is $A_p(\sqrt{p})$ together with the triples with $u=1$ or $a=1$ and $ua\le\sqrt{p}$. Fix $\delta>0$ and split each corner according to whether the relevant product is at most $p^{1/2-\delta}$ or lies in $(p^{1/2-\delta},p^{1/2}]$. The first parts are counted as in Proposition V.2, with the upper bound $|B_p|\le|B_p^{(z)}|$ and the level $(2^{j+1})^{1/2-\delta}$ on the block I_j; they contribute $(a_\delta^{\mathrm{cor}}+o(1))\lambda^2$. For the second parts we drop the condition $\gcd(u,a)=1$ and use the Brun–Titchmarsh inequality, which loses a constant factor but applies to a range of measure $O(\delta)$; they contribute $O(\delta\lambda^2)$ (Appendix A 15). Since δ is arbitrary and $a_\delta^{\mathrm{cor}}\le a_0$, the proposition follows. □

Proof of Theorem B. Let $T_0=\frac12\log N$ and $\sigma=(4a_0/T_0)^{1/3}$. By Propositions IX.1 and VIII.2(i), for N large,

$$\sigma T_0+F^{\sharp}(\sigma)\le\frac32(4a_0)^{1/3}T_0^{2/3}(1+\varepsilon')=(c_0+\varepsilon)(\log N)^{2/3},$$

with ε' chosen in terms of ε. If $Q \le \sqrt{N}$, then $L^\sharp(Q) \le L^\sharp(\sqrt{N}) \le e^{\sigma T_0 + F^\sharp(\sigma)}$, and $(N+Q^2)/L^\sharp(Q) \ge N/L^\sharp(\sqrt{N})$. If $Q > \sqrt{N}$, then by (32) and $\sigma < 2$,

$$\frac{N+Q^2}{L^\sharp(Q)} \ge Q^{2-\sigma} e^{-F^\sharp(\sigma)} \ge N^{1-\sigma/2} e^{-F^\sharp(\sigma)} = N e^{-\sigma T_0 - F^\sharp(\sigma)}.$$

In both cases the bound of the theorem follows. □

X. THE VALUE OF THE CONSTANT

Proposition X.1. *$\mathscr{J} \le \frac{533}{64680}$. Hence $\frac{40}{49} - 6\mathscr{J} + 6\mathscr{K} \ge \frac{1181}{1540}$ and $c_1 \ge c_{\mathrm{var}}$, with $c_{\mathrm{var}}^3 = 286983/(98560\pi^2)$.*

Proof. By the definition (26), $U_\vartheta(s_1, s_2) = 0$ when s_1 or s_2 lies in the window $(2\vartheta, \theta(\vartheta))$, and $U_\vartheta \le 1$ always. On the segment $s_1 + s_2 = \frac{1}{2} + \vartheta$ with $s_1, s_2 > 2\vartheta$, both lie outside the window exactly when $\theta(\vartheta) \le s_1 \le \frac{1}{2} + \vartheta - \theta(\vartheta)$. Hence

$$\ell(\vartheta) \le \Big(\frac{1}{2} + \vartheta - 2\theta(\vartheta)\Big)_+.$$

On $(0, \frac{1}{44}]$ the right side is $\frac{3}{10} + \frac{19}{5}\vartheta$, and on $(\frac{1}{44}, \frac{1}{42})$ it is $39\vartheta - \frac{1}{2}$; both are positive. Integrating,

$$\int_0^{1/44} \Big(\frac{3}{10} + \frac{19\vartheta}{5}\Big) d\vartheta = \frac{151}{19360},$$
$$\int_{1/44}^{1/42} \Big(39\vartheta - \frac{1}{2}\Big) d\vartheta = \frac{251}{569184},$$

with sum $\frac{533}{64680}$. Then $\frac{40}{49} - \frac{6\cdot 533}{64680} = \frac{1181}{1540}$, and since $\mathscr{K} \ge 0$, $c_1^3 \ge \frac{27}{8} \cdot \frac{9}{8\pi^2} \cdot \frac{1181}{1540} = \frac{286983}{98560\pi^2}$. □

The bound of Proposition X.1 uses only (R1). To include (R2) and (R3) we compute $\mathscr{J}$ numerically. For each depth ϑ we solve (25) on a grid of step $\xi/16$, tabulate Dickman's function and Buchstab's function on a grid of step $2 \cdot 10^{-5}$, and evaluate (26) and the two outer integrals of (30) by Gauss–Legendre quadrature, with the nodes split at the points where the integrand has kinks, among them $\vartheta = \frac{1}{54}$ and $\vartheta = \frac{1}{44}$. This gives $\mathscr{J} = 0.000306$, and with $\mathscr{K}$ replaced by 0 in (4) the constant 0.67921, the value that the corners and the strips give without the deep strips. A second program, written independently, solves (25) on the same grid and evaluates the integrals by the midpoint rule; it gives $\mathscr{J} = 0.000300$ and the constant 0.67922, and its loss ℓ agrees with the first to within 0.75% at five depths. A Monte Carlo simulation, which draws the prime factors of u and v by stick breaking and applies the rule of Definition VI.2 directly, agrees with the computed values of U at four points to within 0.002. The value 0.6792 of the corners and the strips is therefore stable to four decimals. The next proposition turns the value of $\mathscr{J}$ into a proved bound.

Proposition X.2. *$\mathscr{J} < 0.000335$. Hence $c_1 > 0.67916$.*

Proof. Appendix A 10 proves $\mathscr{J} < 0.000335$. Since $\pi < 3.141593$ and $\mathscr{K} \ge 0$,

$$c_1^3 \ge \frac{27}{8} \cdot \frac{9}{8\pi^2} \Big(\frac{40}{49} - 6\mathscr{J}\Big)$$
$$> \frac{27}{8} \cdot \frac{9}{8 \cdot 3.141593^2} \Big(\frac{40}{49} - 6 \cdot 0.000335\Big) > 0.67916^3. \ \square$$

The computed bound 0.0003342 lies 9% above the value 0.000306 of the quadrature. The excess has three sources: Lemma A.5 widens the range of r; on each cell the bound takes the least favourable values of ξ, θ and S on the cell; and the grids replace ρ and $\varpi_{\mathscr{A}}$ by step functions that lie above them. Table IV and Figure 4 show that the excess is spread over the depths: on each of the three ranges that carry the integral the bound exceeds the quadrature by 7 to 11%. A finer grid lowers the bound slowly: with the step $\xi_-/32$ for $\varpi_{\mathscr{A}}$ the program gives 0.000345, and with the step $\xi_-/16$, $\tau_k = 64kg$ and about half as many cells it gives 0.000381.

The rules (R1) alone, (R1) with (R2), and (R1) with (R2) and the condition that the part of uv made of primes up to A lies in (A, B), give the losses shown in Figure 5(b) and, without the deep strips, the constants 0.66571, 0.67760 and 0.67831. Almost all the remaining loss of the full rule comes from depths ϑ near $\frac{1}{42}$, where the window of Maynard's theorem closes.

The deep strips are treated in the same way. For the measure $\mathscr{K}$ of (29) a Monte Carlo simulation draws the prime factors of u and v by stick breaking at a fixed depth, places the factors above x^G in the three groups in all ways, and tests the assignments of Definition VII.2 on a grid of targets. The share of the strip that it finds factorable is about 0.74 at the depth $\frac{1}{42}$, about one half at the depth 0.075, and falls to 0 at the depth $\frac{1}{10}$, as Lemma VII.4 requires (Figure 6). Up to the depth 0.045 the values 0.015, 0.02, 0.025, 0.03 and 0.04 of G give shares within 0.03 of each other. Beyond it the smaller values are better, because deeper moduli need finer adjustments of the sizes of their factors: at the depth 0.085 the share for $G = 0.015$ is four times the share for $G = 0.03$. Since Definition VII.2 lets G depend on the pair, the best of these values may be taken at each depth, and then the simulation gives $\mathscr{K} \approx 0.0138$ and $c_1 \approx 0.7015$. It uses 1000 profiles at each depth and each G, the same profiles for every G; with 20,000 profiles at the depths 0.04, 0.06, 0.08 and 0.09 and $G = 0.015$, 0.03 no share moves by more than 0.021. The next proposition proves a lower bound.

Proposition X.3. *$\mathscr{K} > 0.00859$. Hence $c_1 > 0.6931$.*

Proof. Appendix A 11 bounds the density $k(\vartheta)$ of the factorable pairs from below on each cell $[\frac{k}{1024}, \frac{k+1}{1024}]$ with $24 \le k \le 70$, with $G = 0.04$ for $k \le 45$ and $G = 0.03$ for $k \ge 46$. On the first cell only the part above $\frac{1}{42}$ is counted. Since $k(\vartheta) \ge 0$, the depths from $\frac{71}{1024}$ to $\frac{1}{10}$, where the bound is small and its computation long, are counted as 0, and beyond $\frac{1}{10}$ there is nothing to count. The sum of the cell bounds times the lengths exceeds 0.00859 (Table I). Since $\pi < 3.141593$ and

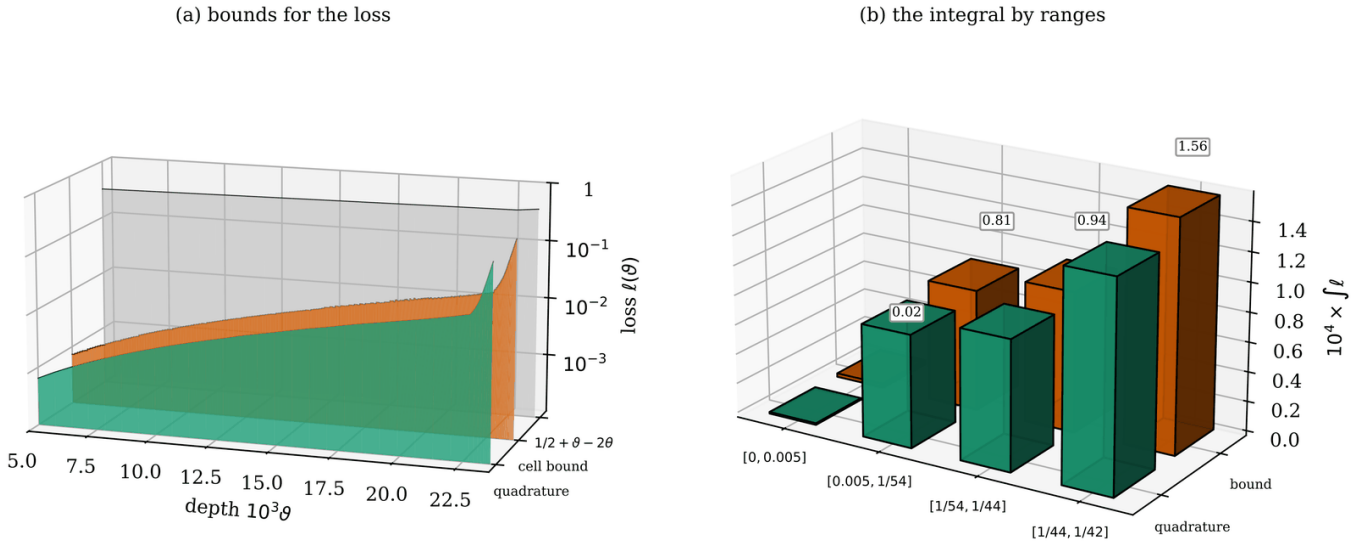


FIG. 4. The certified bound for $\mathscr{J}$. (a) On a logarithmic scale, for $0.005 \le \vartheta < \frac{1}{42}$: the loss $\ell(\vartheta)$ computed by quadrature, the bound on each cell of the computation of Appendix A 10, and the bound $\frac{1}{2}+\vartheta-2\theta(\vartheta)$ of Proposition X.1, which the loss meets at $\vartheta=\frac{1}{42}$, where Maynard's window closes. (b) The integral of the cell bounds and of the quadrature value over the four ranges of Table IV, in units of 10^{-4}.

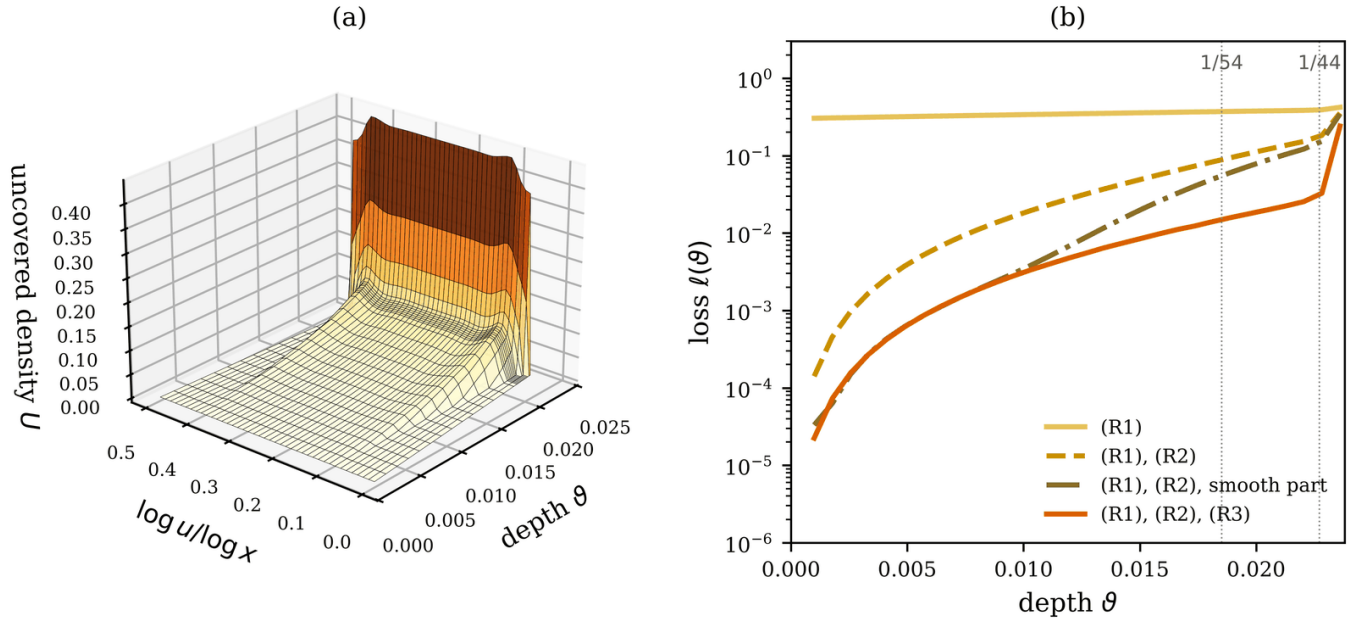


FIG. 5. (a) The density U of the pairs that fail the rule of Definition VI.2, on the segments $s_1+s_2=\frac{1}{2}+\vartheta$, as a function of $s_1=\log u/\log x$ and of the depth ϑ. It vanishes where u or v lies in Maynard's window and is largest near $\vartheta=\frac{1}{42}$. (b) The loss $\ell(\vartheta)$ of (30) for the rule (R1) alone, for (R1) with (R2), for an intermediate rule that also accepts the part of uv made of primes up to A when it lies in the window, and for the full rule (R1)–(R3). The dotted lines mark $\vartheta=\frac{1}{54}$ and $\vartheta=\frac{1}{44}$, where the window changes its shape.

$\mathscr{J} < 0.000335$ (Proposition X.2),

$$c_1^3 > \frac{27}{8}\cdot\frac{9}{8\cdot 3.141593^2} \times\Big(\frac{40}{49}-6\cdot 0.000335+6\cdot 0.00859\Big) > 0.6931^3. \quad \square$$

The bound captures 62% of the simulated value. The difference has three sources. The certificate stops at the depth $\frac{71}{1024}$, beyond which the simulation finds 0.0025 of $\mathscr{K}$. It uses one value of G on each cell, while the simulation takes the best of five at each depth, and the deeper cells lose most here: at the depth 0.08 the value $G=0.03$ gives less than half of the share of $G=0.015$. And it counts only the configurations that it decides within seven prime factors above x^G and at the smallest width of its tree; the configurations that fail at that width carry

TABLE I. The certified lower bound for $\mathscr{K}$ and the Monte Carlo value by ranges of the depth. The cells have length $\frac{1}{1024}$; the first range starts at $\frac{1}{42}$, and the last row, from $\frac{71}{1024}$ to $\frac{1}{10}$, is not certified.

Depths	Cells	G	Bound	Monte Carlo
0.0238–0.0352	24–35	0.04	0.00296	0.0034
0.0352–0.0449	36–45	0.04	0.00219	0.0026
0.0449–0.0566	46–57	0.03	0.00207	0.0028
0.0566–0.0693	58–70	0.03	0.00137	0.0025
0.0693–0.1000			0	0.0025
All	24–70		0.00859	0.0138

TABLE II. The constant given by the large sieve for regions of the triangle of Figure 1. The density is $a =$ (share of the triangle)$\cdot\kappa/4$, and $c = \frac{3}{2}a^{1/3}$. The rows with strips count them through the rules of Definition VI.2; the row with all of the strips would need a divisor in Maynard's range for every modulus. The rows with deep strips add $6\mathscr{K}$, with the certified bound and with the Monte Carlo value of $\mathscr{K}$; the row to the depth $\frac{1}{10}$ would need every modulus of the strips and deep strips to split, and the last row would need moduli beyond $p^{3/5}$.

Region	Share	a	c
One corner	0.25	0.0284966	0.4582
Two corners	0.5	0.0569932	0.5773
Three corners	0.75	0.0854897	0.6608
Strips by (R1)	0.766883	0.0874142	0.6657
Strips by (R1), (R2)	0.808696	0.0921802	0.6776
Strips by (R1)–(R3)	0.814491	0.0928408	0.6792
All of the strips	0.816327	0.0930501	0.6797
With the deep strips, certified	0.866031	0.0987157	0.6932
With the deep strips, Monte Carlo	0.897213	0.1022700	0.7015
To the depth $\frac{1}{10}$	0.96	0.1094269	0.7175
Whole triangle	1	0.1139863	0.7273

most of the rest.

XI. REMARKS

Remark XI.1 (Smaller and larger regions)**.** The argument applies to any union of corners, and the density is proportional to the area covered (Table II and Figure 7). A single corner gives $c = \frac{3}{2}(\kappa/16)^{1/3} = 0.4582$, two corners give $\frac{3}{2}(\kappa/8)^{1/3} = 0.5773$, and the three corners give c_0. Each row with strips includes the three corners; the share of the strips counted by (R1) alone is the closed form $\frac{1181}{1540}$ of Proposition X.1, and the share of all of the strips is $\frac{40}{49}$. The deep strips below the depth $\frac{1}{10}$ have area $\frac{88}{3675}$, so that every pair counted up to that depth would give the share $\frac{40}{49} + \frac{6\cdot 88}{3675} = \frac{24}{25}$; the simulation finds a little more than half of these pairs factorable.

Remark XI.2 (The level)**.** Section III uses the level $16X^2 \le p$. The factor 16 does not affect the constants, since only $\log X/\log p \to \frac{1}{2}$ enters the density, but the bounds (15) do depend on it. For the primes $p \equiv 3 \pmod 4$ below 10^6 no two triples in the corners at level $\sqrt{p}/4$ share a class, while at level $\sqrt{p}/2$ there are 17,568 such pairs, all between B and C. The first is at $p = 31$, where $(1,4,2) \in B_{31}(2)$ and $(8,1,1) \in C_{31}(2)$ both have the class 8; here $P = 8$, far below the bound $P \ge 1024$ of (15).

Remark XI.3 (Where the method stops)**.** The strips have depth $\frac{1}{42}$ because the range of divisors in Theorem VI.1 is empty for $\vartheta \ge \frac{1}{42}$. If every triple of the strips were counted, the share would be $\frac{3}{4} + 6(\frac{1}{84} - \frac{3}{3528}) = \frac{40}{49}$ and the constant $\frac{3}{2}(45/(49\pi^2))^{1/3} = 0.6797$; the rule of Definition VI.2 loses $6\mathscr{J} = 0.0018$ of this share. Beyond the depth $\frac{1}{42}$ Theorem VII.1 counts the factorable pairs, and Lemma VII.4 stops it at the depth $\frac{1}{10}$, which is the level $p^{3/5}$ of Maynard's second theorem: the second and third conditions of (27) ask for a factor of the modulus above $p^{2\vartheta}$ and below $p^{1/2-3\vartheta}$. Even if every pair up to that depth were counted, the share would be $\frac{24}{25}$ and the constant $\frac{3}{2}(27/(25\pi^2))^{1/3} = 0.7175$. A larger constant needs primes in a fixed residue class to moduli beyond $p^{3/5}$. Under the Elliott–Halberstam conjecture [10] the whole centre could be counted through the smallest of the three products, which is at most $P^{2/3}$; since the classes of the centre are distinct and differ from those of the corners apart from the pairs counted by $\nu'(p)$ (Propositions IV.3 and IV.4), the whole triangle would then give $\frac{3}{2}(\kappa/4)^{1/3} = 0.7273$.

Remark XI.4 (Vaughan's count of classes)**.** In the form given in [23, Section 4], Vaughan's argument uses at each prime $f(p) = \lfloor \frac{1}{2}\sum_{t\mid P}|\mu(t)|\tau(P/t)\rfloor$ classes. The sum equals $\sum_{d\mid P} 2^{\omega(d)}$, the number of Type II triples, so $f(p)$ is half of $|\mathscr{T}(p)|$. If the primes are equidistributed in progressions to moduli up to $p^{2/3}$, the number of triples has density $\kappa/4$, the whole triangle, and the classes $f(p)$ have density at most $\kappa/8$. By Proposition VIII.2(i) and the argument of Theorem B, no sieve with $f(p)$ classes at each prime can then give a constant above $\frac{3}{2}(\kappa/8)^{1/3} = 0.5773$, below both c_0 and c_1.

XII. FORMAL AND NUMERICAL VERIFICATION

The algebraic steps are checked in Lean 4 [13] with Mathlib [15], in 59 theorems. For the corners: the normal forms (8); the identities (9) and (10); the identities (17), (11), (12) and (14) as consequences of their hypotheses; the three inequalities (15); the relation $\varphi(qn) = q\,\varphi(n)$ for $q \mid n$; the local factors of S_B; the optimisation in Proposition VIII.2; and the value $c_0^3 = 729/(256\pi^2)$. For the centre: the Farey bound and the bound $4u^2\alpha < p+1$ of Proposition IV.3; the identities (20)–(22), the inequality $4ks \le a$ and the case $j = 0$ of Proposition IV.4; and the family of Remark IV.6, with the primality of 531359 and 1442783. For the strips: the range of Theorem VI.1 and the values $\frac{1}{44}$ and $\frac{1}{54}$; the divisor chain of Lemma VI.3; the area $\frac{13}{1176}$ and the share $\frac{40}{49}$. For Proposition X.2: the monotonicity of the bound of Proposition X.1, the lower bound for ξ on a cell, the geometric sum behind (A4), and the final inequality of the proof, with $\pi < 3.141593$ from Mathlib. For the deep strips: the bound $5s < 3$ behind

(a) the density k: $\mathcal{K} > 0.00859$ certified

(b) Monte Carlo share by depth and G

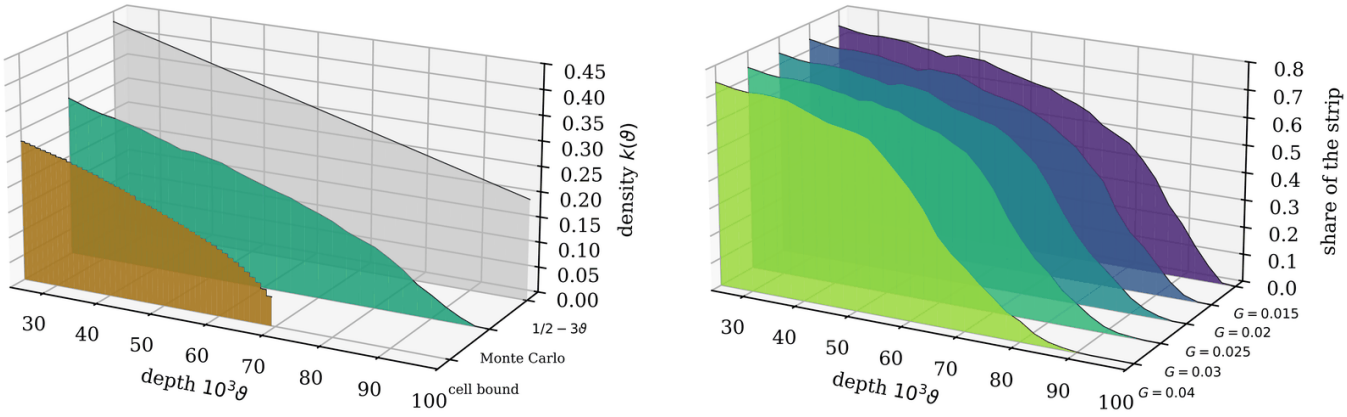


FIG. 6. The deep strips. (a) For $\frac{1}{42} \le \vartheta \le \frac{1}{10}$: the certified lower bound for the density $k(\vartheta)$ of the factorable pairs on each cell of the computation of Appendix A 11, the density given by the Monte Carlo simulation with the best G at each depth, and the length $\frac{1}{2} - 3\vartheta$ of the strip, which bounds k. Their integrals are $\mathcal{K} > 0.00859$, $\mathcal{K} \approx 0.0138$ and $\frac{88}{3675}$. (b) The share of the strip that the simulation finds factorable, by depth and G. Smaller values of G are better at larger depths, and every share vanishes at $\vartheta = \frac{1}{10}$ (Lemma VII.4).

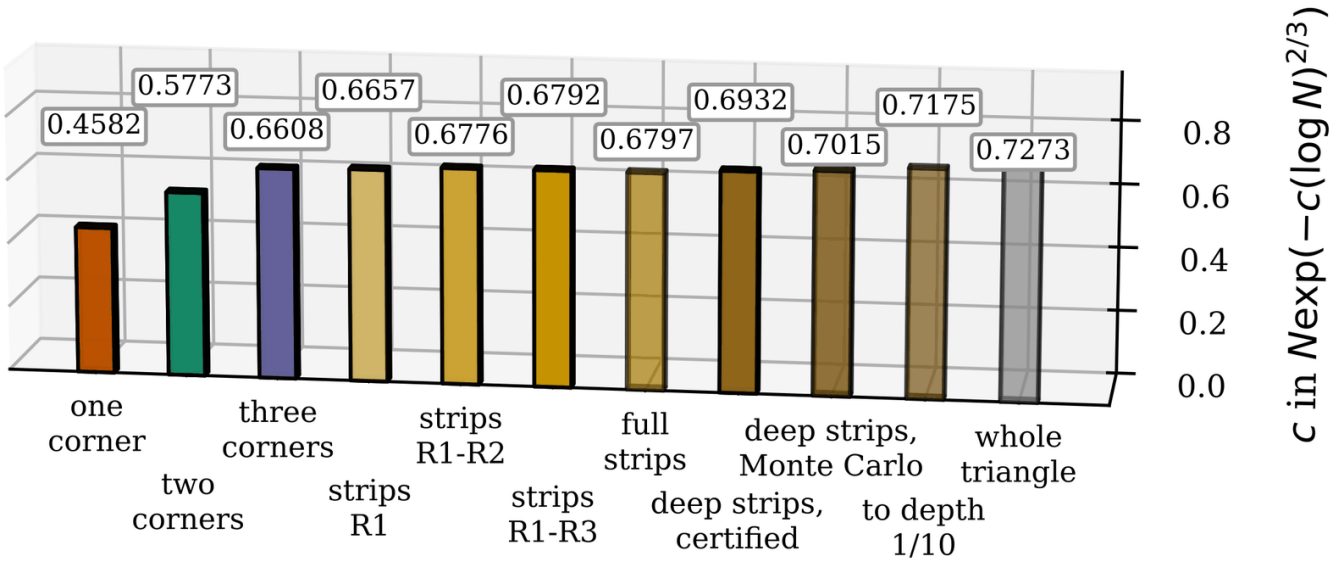


FIG. 7. The constant c in $N\exp(-c(\log N)^{2/3})$ for one, two and three corners, for the corners with the strips counted by the three rules, for the corners with all of the strips, with the deep strips (certified bound and Monte Carlo value), with every pair counted to the depth $\frac{1}{10}$, and for the whole triangle. The last two are not available.

Lemma VII.4, the monotonicity of the conditions (27), the areas $\frac{3}{98}$ and $\frac{88}{3675}$, the share $\frac{24}{25}$, and the final inequality of Proposition X.3. All these theorems depend at most on the axioms `propext`, `Classical.choice` and `Quot.sound`. The analytic estimates of Sections V, VI and VIII are not formalised.

Independent programs in C, C++, Python and Julia check the remaining steps numerically. Over the 332,398 primes $p \equiv 3 \pmod 4$ below 10^7, with 37,093,810 Type II triples, the corners at level $\sqrt{p}/16$ are disjoint, the three normal forms agree, the classes in each corner are distinct, and there is no coincidence at all between different corners. The same primes have 6,943,856 triples in the centre, of which 1,760,868 have their largest coordinate at least $P^{1/2-1/42}$. The classes of the centre are distinct; they agree with no class of A at level $\sqrt{p}/4$, with no class of B for $u < a$ and no class of C for $u > a$; and the remaining coincidences are the four pairs of Remark IV.6. The divisor chain of Lemma VI.3 holds in all 128,000,000 cases of the integers up to $2 \cdot 10^6$ with 64 windows. The sums

$\sum_p |A_p|/p$ and $\sum_p |B_p|/p$ agree with the predictions $\sum_p S_A/p$ and $\sum_p S_B/p$ to within 0.25%. The parametrisation of Proposition III.2 is checked by brute force against all coincidences with $P \le 4000$, without the restriction to the corners. The mean values of Lemma V.1 are computed to $X = 2\cdot 10^6$, where $(S - \frac{\kappa}{2}\log^2 X)/\log X$ settles near -0.47 for S_A and 0.55 for S_B. The two bounds of Proposition VIII.2 are evaluated for the density a_0 up to $T = 10^{15}$, where their ratio to the exact optimum is 1.0000. The integral $\mathscr{J}$ is computed as in Section X, and the bound of Proposition X.2 by the C program of Appendix A 10, which runs in about 25 minutes on one processor core. For the deep strips a C program simulates the density of the factorable pairs at a given depth and G, and the bound of Proposition X.3 is computed cell by cell by the C program of Appendix A 11, in about 5 hours on one processor core in all; a Python script adds up the cell bounds in exact rational arithmetic.

The engine runs. In the first run the library of Anaxia held 22 typed construction nodes: the large sieve with the tilted lower bound; each corner paired with each of the three normal forms, or left out; the centre, left out or counted under the Elliott–Halberstam conjecture; and the inputs that each of these needs. Typed crossing produced 128 closed constructions, which the engine tested on the 39,322 primes $p \equiv 3 \pmod 4$ below 10^6. It kept 7, those with one, two or three corners and the matching normal forms; it rejected 112 because a normal form failed to separate the classes, 8 that contain the centre because a class of the centre equals a class of a corner at $p = 531359$ (Remark IV.6), and the empty construction. The best of the seven gives c_0. In the second run the library held 19 nodes for the centre: the three normal forms; five rules for finding a divisor of the modulus in Maynard's range, from (R1) alone to the rule of Definition VI.2, and a rule that omits the bound on the steps of the chain; the family of coincidences of Proposition IV.4, allowed or not; and the Elliott–Halberstam conjecture. Of its 37 closed constructions it accepted 5, with the constants 0.660789 (centre left out), 0.665710, 0.677596, 0.678311 and 0.679210, marked one as conditional (the whole centre under the Elliott–Halberstam conjecture, 0.727292), and rejected 31: 24 because the forms $4u^2\alpha$ and $(4a^2\alpha)^{-1}$ do not separate the classes of the centre (they fail on all 12,399 primes tested), 6 because they do not allow the two coincidences of the family below 10^6, and the rule without the bound on the steps, which claims a divisor that does not exist 1,449,392 times among the integers up to $2\cdot 10^5$ and 64 windows, first at $n = 4$. The best construction is the one of this paper. In the third run, with version 3.1.0 [2], the library held 17 nodes for the deep strips: the deep strips left out, counted by Maynard's window continued past $\frac{1}{42}$, split for Theorem VII.1, or counted under the Elliott–Halberstam conjecture; three splitting rules, which fill one group, two groups with the reserve G of (28), or two groups without it; four sets of values of G, namely $\{0.04\}$, $\{0.03\}$, $\{0.02\}$ and the best of $\{0.015, 0.02, 0.025, 0.03, 0.04\}$ at each depth; the two theorems of Maynard and the Elliott–Halberstam conjecture; and the sieve, the corners and the strips of the first two runs. Of its 15 closed constructions it accepted 9, marked one as conditional (the deep strips under the Elliott–Halberstam conjecture, 0.727292), and rejected 5: the window, which is empty at every depth from $\frac{1}{42}$ on, and the four rules without a reserve. Of the 89,890 to 133,334 assignments that each of these rules allows and that the engine tested on the integers from 10^9 to $10^9 + 10^5$, between 25,404 and 35,558 could not be brought to their targets by the fill. The accepted constructions leave the deep strips out (0.679210) or split them with one or two filled groups; with the best G at each depth both rules give $\mathscr{K} \approx 0.0138$ and $c_1 \approx 0.7015$ in the simulation, and the rule with two filled groups, which contains the rule with one, is that of Definition VII.2. The records of the three runs and the scripts that reproduce them are in the archive [28].

Appendix A: Detailed derivations

1. The local factors

The totient of $4n$. If n is odd then $\varphi(4n) = \varphi(4)\varphi(n) = 2\varphi(n)$. If $n = 2^\beta n'$ with $\beta \ge 1$ and n' odd then $\varphi(4n) = 2^{\beta+1}\varphi(n') = 4\varphi(n)$. In both cases

$$\frac{1}{\varphi(4n)} = \frac{\psi(n)}{2n}, \qquad \psi(n) = \prod_{q\mid n,\ q \text{ odd}} \frac{q}{q-1}. \tag{A1}$$

For a prime q dividing n, $\varphi(qn) = q\,\varphi(n)$.

The corner B. With (A1),

$$S_B(X) = \frac{1}{2}\sum_{u\alpha\le X} \frac{W_B(u,\alpha)}{u\alpha}, \qquad W_B(u,\alpha) = \frac{\varphi(u)}{u}\,\psi(u\alpha).$$

The function W_B is multiplicative in the pair: $W_B(u,\alpha) = \prod_q W_B(q^i, q^j)$ with $q^i \parallel u$ and $q^j \parallel \alpha$. At an odd prime q its local values are $W_B = 1$ at $(0,0)$, $W_B = (1-\frac{1}{q})\frac{q}{q-1} = 1$ for $i \ge 1$, and $W_B = \frac{q}{q-1}$ for $i = 0$, $j \ge 1$. At $q = 2$ the factor ψ is absent and $W_B = \frac{1}{2}$ for $i \ge 1$, $W_B = 1$ for $i = 0$.

The local generating function at an odd prime q is

$$\sum_{i,j\ge 0} W_B(q^i,q^j)x^i y^j = 1 + \frac{x}{(1-x)(1-y)} + \frac{q}{q-1}\cdot\frac{y}{1-y},$$

and multiplying by $(1-x)(1-y)$ gives

$$G_q(x,y) = 1 + \frac{y}{q-1} - \frac{xy}{q-1}.$$

At $q = 2$ the same computation gives $G_2(x,y) = 1 - \frac{x}{2}$. Hence $W_B(u,\alpha) = \sum_{d\mid u,\, e\mid\alpha} g(d,e)$, where g is multiplicative in the pair with the local values read off from G_q: $g(1,q) = \frac{1}{q-1}$ and $g(q,q) = -\frac{1}{q-1}$ for odd q, $g(2,1) = -\frac{1}{2}$, and $g = 0$ at all other prime powers except $g(1,1) = 1$. At $x = y = 1/q$,

$$G_q\Big(\frac{1}{q},\frac{1}{q}\Big) = 1 + \frac{1}{q(q-1)} - \frac{1}{q^2(q-1)} = 1 + \frac{1}{q^2},$$

and $G_2(\frac{1}{2},\frac{1}{2}) = \frac{3}{4}$. The same value $1 + q^{-2}$ is the quotient of the local sum $\sum_{i,j} W_B(q^i,q^j)q^{-i-j} = (q^2+1)/(q-1)^2$ by $(1-1/q)^{-2}$.

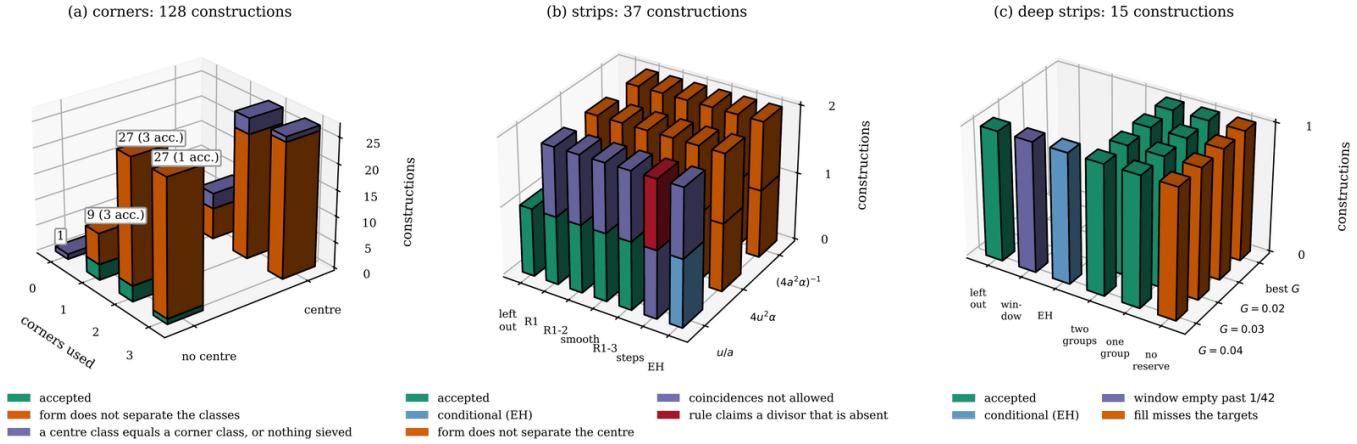


FIG. 8. The three engine runs. (a) The 128 constructions of the corners, by the number of corners and by whether the centre is included. Seven are accepted; the rest are rejected because a normal form does not separate the classes, because a class of the centre equals a class of a corner, or because they sieve nothing. (b) The 37 constructions of the strips, by the rule that finds the divisor and by the normal form used in the centre, with the family of coincidences allowed (front) or not (back). Five are accepted, and the whole centre under the Elliott–Halberstam conjecture is conditional. (c) The 15 constructions of the deep strips, by the rule that splits the moduli and the values of G it may use. The window of Theorem VI.1 is empty there, the rules without a reserve miss their targets, and the deep strips under the Elliott–Halberstam conjecture are conditional.

The corner A. Here

$$S_A(X)+R(X)=\frac{1}{2}\sum_{ua\le X}\frac{W_A(u,a)}{ua},$$

where $W_A(u,a)=\psi(ua)$ when $\gcd(u,a)=1$, $W_A=0$ otherwise, and $R(X)$ is the sum over the pairs with $u=1$ or $a=1$. Since $\sum_{n\le X}1/\varphi(4n)=O(\log X)$, $R(X)=O(\log X)$. The local values of W_A at an odd prime are 1 at $(0,0)$, $\frac{q}{q-1}$ when exactly one exponent is positive, and 0 when both are; at $q=2$ they are $1,1,0$. The local generating functions after multiplication by $(1-x)(1-y)$ are

$$G_q^A(x,y)=1+\frac{x+y}{q-1}-\frac{(q+1)xy}{q-1},\qquad G_2^A=1-xy,$$

with $G_q^A(\frac{1}{q},\frac{1}{q})=1+q^{-2}$ and $G_2^A(\frac{1}{2},\frac{1}{2})=\frac{3}{4}$. The coefficients give $W_A=\sum_{d|u,e|a}h_A(d,e)$ with $\sum_{d,e}|h_A(d,e)|/(de)<\infty$, as Lemma VI.4 requires, and the same holds for g.

TABLE III. The local factors of the two mean values. Both corners give $1+q^{-2}$ at odd q and $\frac{3}{4}$ at 2.

	$G(x,y)$ at odd q	at $x=y=\frac{1}{q}$	$q=2$
B	$1+\frac{y-xy}{q-1}$	$1+q^{-2}$	$\frac{3}{4}$
A	$1+\frac{x+y-(q+1)xy}{q-1}$	$1+q^{-2}$	$\frac{3}{4}$

The constant. The Euler product of the values at $x=y=1/q$ is

$$\frac{3}{4}\prod_{q\text{ odd}}\Big(1+\frac{1}{q^2}\Big)=\frac{3}{4}\cdot\frac{4}{5}\cdot\frac{\zeta(2)}{\zeta(4)}=\frac{3}{4}\cdot\frac{4}{5}\cdot\frac{15}{\pi^2}=\frac{9}{\pi^2}=2\kappa,$$

using $\zeta(2)=\pi^2/6$ and $\zeta(4)=\pi^4/90$.

2. The hyperbola sum

For $Y\ge 1$ let $H(Y)=\sum_{mn\le Y}1/(mn)$ and $H(Y)=0$ for $Y<1$. With $h(x)=\sum_{m\le x}1/m=\log x+\gamma+O(1/x)$,

$$H(Y)=\sum_{m\le Y}\frac{1}{m}h\Big(\frac{Y}{m}\Big)=\sum_{m\le Y}\frac{\log Y-\log m+\gamma}{m}+O\Big(\frac{\log Y}{Y}\Big).$$

Since $\sum_{m\le Y}(\log m)/m=\frac{1}{2}\log^2 Y+O(1)$, this gives $H(Y)=\frac{1}{2}\log^2 Y+2\gamma\log Y+O(1)$.

Write $\Lambda=\log X$. Then, with g as in Appendix A 1,

$$2S_B(X)=\sum_{de\le X}\frac{g(d,e)}{de}\Big(\frac{1}{2}(\Lambda-\log de)^2+O(1+\Lambda)\Big).$$

The series $\sum_{d,e}|g(d,e)|(1+\log de)^2/(de)$ converges, since its Euler factor at an odd q is $1+O(q^{-2}\log^2 q)$. For the pairs with $de>X$, $\sum_{de>X}|g(d,e)|/(de)\le X^{-1/4}\sum_{d,e}|g(d,e)|(de)^{-3/4}$, and the last series converges because its terms at q are $O(q^{-7/4})$ and $O(q^{-5/2})$. Expanding the square,

$$2S_B(X)=\frac{\Lambda^2}{2}\sum_{d,e}\frac{g(d,e)}{de}+O(\Lambda)=\kappa\Lambda^2+O(\Lambda).$$

For S_A the coefficients of G_q^A are of size q^{-1} at $(1,0)$ and $(0,1)$ and of size 1 at $(1,1)$, so the terms of $\sum|h_A(d,e)|(de)^{-3/4}$ at q are $O(q^{-7/4})$ and $O(q^{-3/2})$; the rest is identical.

3. The Bombieri–Vinogradov theorem with multiplicities

The Bombieri–Vinogradov theorem [6], [8, Chapter 28] states that for every $A>0$ there is $B>0$ such that

$$\sum_{q\le x^{1/2}(\log x)^{-B}} \max_{(b,q)=1} \max_{y\le x} \Big|\pi(y;q,b)-\frac{\pi(y)}{\varphi(q)}\Big| \ll_A \frac{x}{(\log x)^A}.$$

Let $E(q)$ be the inner maximum. In the proof of Proposition V.2 each modulus $q=4n$ occurs with multiplicity at most $\tau_3(n)$. Using the trivial bound $E(q)\ll x/q$, Cauchy's inequality and $\sum_{n\le x}\tau_3(n)^2/n\ll(\log x)^9$,

$$\sum_q \tau_3(q)E(q) \le \Big(\sum_q \tau_3(q)^2E(q)\Big)^{1/2}\Big(\sum_q E(q)\Big)^{1/2} \ll \frac{x}{(\log x)^{(A-9)/2}}.$$

Taking $A=15$ gives the error $O(x(\log x)^{-3})$ used in the proof. The inclusion–exclusion over $d\mid(u,P_z)$ replaces each pair by the progressions modulo $4u\alpha d$ with $d\mid u$; a modulus $4m$ arises from at most $\tau_3(m)$ triples (u,α,d), so the same estimate covers all of them. The moduli $4u\alpha d\le 4X_je^{2z}$, since $\prod_{q\le z}q\le e^{2z}$, are below $x^{1/2}(\log x)^{-B}$ for j large because $X_j=2^{j(1/2-\delta)}$ and $z=j^{1/2}$.

4. Maynard's theorem with multiplicities

In the proof of Proposition VI.7, fix $y\in[x,2x]$ and let $E(q;y)=|\pi(y;q,-1)-\pi(y)/\varphi(q)|$. For $q\le y$ we have $E(q;y)\le\pi(y;q,-1)+\pi(y)/\varphi(q)\ll y\log\log y/q$, and each modulus q that occurs lies in $\mathcal{Q}_{\vartheta_i^+,\eta}(y)$ and occurs at most $\tau_3(q/4)$ times. Since $\sum_{n\le y}\tau_3(n)^2/n\ll(\log y)^9$, the sum of $\tau_3(q/4)^2E(q;y)$ over these moduli is $O(y(\log y)^{10})$, and by Theorem VI.1 with $A=17$ the sum of $E(q;y)$ is $O(y(\log y)^{-17})$. Cauchy's inequality gives

$$\sum_q \tau_3(q/4)E(q;y)\ll\frac{y}{(\log y)^{7/2}}.$$

Let $f(p)=|\mathscr{Z}^A_{p,i}|$ and $F(y)=\sum_{x<p\le y}f(p)$, and let S be the sum of $1/\varphi(4ua)$ over the pairs (u,a) of the proof of Proposition VI.7. Applying the bound at y and at x gives $F(y)=(\pi(y)-\pi(x))S+O(x(\log x)^{-7/2})$ at each of the $j+1$ points $y_l=x(1+l/j)$. On $(y_l,y_{l+1}]$ the weight $1/p$ lies between $1/y_{l+1}$ and $1/y_l$, whose ratio is at most $1+1/j$, and $f\ge0$, so

$$\sum_{y_l<p\le y_{l+1}}\frac{f(p)}{p}=\frac{F(y_{l+1})-F(y_l)}{y_l}\Big(1+O\Big(\frac1j\Big)\Big).$$

The same holds with f replaced by S times the indicator of the primes. Summing over l,

$$\sum_{p\in I_j}\frac{f(p)}{p}=S\sum_{p\in I_j}\frac1p\Big(1+O\Big(\frac1j\Big)\Big)+O\big(j(\log x)^{-7/2}\big).$$

Since $S\ll j^2$ and $\sum_{p\in I_j}1/p\ll1/j$, the error is $O(1)$. The piece B is treated in the same way with the moduli $4u\alpha d$.

5. Proof of Theorem VII.1

We follow the proof of [17, Theorem 1.1] in [17, Section 5]; the changes concern the range of the Type II variable, the form of the weights and the moduli with a large square factor. Write $\mathscr{L}=\log y$ and $z_0=y^{1/(\log\log y)^3}$, and recall that a sequence α satisfies the Siegel–Walfisz condition if, for all $d,q\ge1$, $(b,q)=1$, $A>0$ and $N\ge2$,

$$\sum_{\substack{n\sim N\\ n\equiv b(q)\\ (n,d)=1}}\alpha_n-\frac{1}{\varphi(q)}\sum_{\substack{n\sim N\\ (n,dq)=1}}\alpha_n\ll_A\frac{N\tau(d)^{O(1)}}{(\log N)^A},\tag{A2}$$

where $n\sim N$ means $N<n\le2N$. The sequences $\mathbf{1}_{P^-(n)\ge z_0}$, $\mathbf{1}_{P^-(n)\ge z_0}\log n$ and $\mathbf{1}_{P^-(n)\ge z_0}\mu(n)$ satisfy it, as used in [17, Section 5].

The moduli with a square factor. Let $E=\lambda-\sum_j\gamma^{(j)}*\beta^{(j)}*\nu^{(j)}$ for one value of n. For $q\le y^{2/3}$ the trivial bound and Shiu's theorem [17, Lemma 6.1] give

$$\sum_{\substack{m\le2y\\ m\equiv-1(q)}}\tau_6(m)^C\ll\frac{y(\tau(q)\mathscr{L})^{O_C(1)}}{q},$$

so

$$\sum_q|E_q|\sum_{\substack{m\le2y\\ m\equiv-1(q)}}\tau_6(m)^C \ll y\mathscr{L}^{O(1)}\sum_{p'>y^\eta}\sum_{q'}\frac{\tau(p'^2q')^{O(1)}}{p'^2q'}\ll y^{1-\eta/2},$$

and the same holds with the progression replaced by $1/\varphi(q)$ times the sum over $m\le2y$. Every sum below has this form, with coefficients bounded by $\tau_6(m)^C$, so in it λ may be replaced by the convolutions attached to the relevant n.

Bounded coefficients. As in the proofs of [17, Propositions 5.1 and 5.2], Shiu's theorem shows that the moduli q with $\tau(q)\ge\mathscr{L}^C$, and the integers m at which one of the sequences below exceeds $\mathscr{L}^C$, contribute $O(y\mathscr{L}^{O(1)-C})$. Restricting each of $\gamma^{(j)},\beta^{(j)},\nu^{(j)}$ to $\tau<\mathscr{L}^C$ changes the convolutions only at moduli with $\tau(q)\ge\mathscr{L}^C$, and dividing by $\mathscr{L}^{3BC}$ we may assume these sequences 1-bounded. A Siegel–Walfisz sequence truncated to $\tau(n)<\mathscr{L}^C$ still satisfies (A2) [17, Lemma 6.5].

The decomposition. By partial summation, the negligible contribution of prime powers and the condition $P^-(n)\ge z_0$ for primes $n>z_0$, it suffices to show for every $t\in[y,2y]$ that

$$\sum_q\lambda_q\sum_{\substack{y<n\le t\\ P^-(n)\ge z_0}}\Lambda(n)\Big(\mathbf{1}_{n\equiv-1(q)}-\frac{\mathbf{1}_{(n,q)=1}}{\varphi(q)}\Big)\ll\frac{y}{\mathscr{L}^A}.$$

The identity of Heath-Brown with $k=3$ [17, Lemma 5.3] writes $\Lambda(n)$ as a combination of convolutions of six variables n_1,n_2,n_3,m_1,m_2,m_3, with the weights $\log n_1$, 1, 1, $\mu(m_1)$, $\mu(m_2)$, $\mu(m_3)$, all supported on $P^-\ge z_0$ and with $m_i\le(2y)^{1/3}$. We put each variable in a dyadic range, $n_i\sim N_i$ and

$m_i \sim M_i$, which gives $O(\mathscr{L}^6)$ sums, and write $e_v = \log V/\log y$ for the range V of a variable v. Put

$$a = \tfrac{1}{6} - \vartheta - 3\varepsilon, \qquad b = \tfrac{1}{3} + 3\varepsilon,$$

and note that $2a < b$.

Case (i): some set S of the six variables has $\sum_{v\in S} e_v \in [a,b]$. By [17, Lemma 6.2] the condition $y < n \le t$ can be replaced by conditions $v \in I_v$ on the variables, with intervals $I_v \subset [V,2V]$, at the cost of a factor $\mathscr{L}^{O(1)}$ and an admissible error. Let α be the convolution of the variables in S and β' that of the others. The support of α lies in $(\prod_S V, 2^6\prod_S V]$; we split it into at most six dyadic ranges $n \sim N$, and then $\log N/\log y \in \mathscr{N}_\vartheta$ for y large. Each restriction $\alpha^{(N)}$ of α to such a range satisfies (A2). Indeed, let $v_0 \in S$ have the largest range V_0, so that $V_0 \ge N^{1/6}/2$, and write $n = n'v_0$ with n' the product of the other variables of S. The conditions $n \equiv b \pmod q$ and $(n,d)=1$ become $(n',dq)=1$, $(v_0,d)=1$ and $v_0 \equiv b\bar{n}' \pmod q$, and v_0 runs over the interval $I_{v_0} \cap (N/n', 2N/n']$. For a sequence w satisfying (A2) and an interval $(c_1,c_2] \subset [V_0,4V_0]$, we have $\mathbf{1}_{(c_1,c_2]} = \mathbf{1}_{(0,c_2]} - \mathbf{1}_{(0,c_1]}$ and $\mathbf{1}_{(0,c]} = \sum_{k\ge1} \mathbf{1}_{(c/2^k, c/2^{k-1}]}$; the ranges with $c/2^k \ge V_0^{1/2}$ satisfy (A2), and the rest has total $O(V_0^{1/2}\log V_0)$. Summing over n', the left side of (A2) for $\alpha^{(N)}$ is $\ll N\mathscr{L}^{O(1)}\tau(d)^{O(1)}(\log V_0)^{-A}$, and $\log V_0 \ge \frac{1}{6}\log N - 1$. The same argument applies to the restrictions of $\alpha^{(N)}$ to subintervals of $(N,2N]$. Now we use the hypothesis at $n = \log N/\log y$: $\lambda = \sum_j \gamma^{(j)} * \beta^{(j)} * \nu^{(j)}$ up to E, with supports $q_i \le y^{d_i}$. Putting q_1,q_2,q_3 into dyadic ranges $Q_i' < y^{d_i}$, the admissibility with margin 15ε gives the four conditions of [17, Proposition 8.2] with ε, because they become weaker as the Q_i' decrease, and that proposition, applied to $\alpha^{(N)}$ and β', bounds the sum.

Case (ii): there is no such set. Call a variable small if $e_v < a$ and large if $e_v > b$; every variable is one or the other. The product of the small variables has exponent below a: otherwise, adding them one at a time, the partial sums of the e_v would pass from below a to a value in $[a,2a) \subset [a,b]$. Hence the large variables have exponents adding up to more than $1-a-o(1) > b$, so there are one or two of them, since three would exceed 1, and they are among n_1,n_2,n_3 because $e_{m_i} \le \frac{1}{3} + o(1) < b$. If there is one, it runs over an interval of length at least $y^{1-a}/64 \ge q y^{\varepsilon}$ for $q \le y^{1/2+\vartheta}$, since $\frac{1}{2}+\vartheta+\varepsilon < \frac{5}{6}+\vartheta+3\varepsilon = 1-a$, and [17, Lemma 5.4], summed over the other variables and over q with $|\lambda_q| \le 1$, bounds the sum. If there are two, they have ranges $N_1, N_2 \ge y^{1/3} \ge y^{3\varepsilon}$, the product M of the ranges of the others satisfies $M < 64 y^a$, and

$$y^{1/2+\vartheta} = \frac{y^{2/3-3\varepsilon}}{y^a} < \frac{(2y)^{2/3-3\varepsilon}}{M}$$

for y large, so [17, Proposition 5.2], which bounds the sum over q of the absolute values and allows the weight $\log n_1$, bounds the sum. There is no case without a large variable, because the exponents of all six add up to $1+o(1)$.

6. Large prime factors of $\gcd(u,a)$

The truncated product. For $u \ge 1$,

$$0 \le \prod_{\substack{q\mid u\\ q\le z}} \Big(1-\frac{1}{q}\Big) - \frac{\varphi(u)}{u} \le \sum_{\substack{q\mid u\\ q>z}} \frac{1}{q}.$$

Writing $u = qu'$ and using $\varphi(4qu'\alpha) \ge (q-1)\varphi(4u'\alpha)$,

$$S_B^{(z)}(X) - S_B(X) \le \sum_{q>z} \frac{1}{q(q-1)} \sum_{u'\alpha\le X} \frac{1}{\varphi(4u'\alpha)} \ll \frac{\log^2 X}{z},$$

because $\sum_{n\le X} \tau(n)/\varphi(4n) \ll \log^2 X$. The same bound, with $X = x^K$, controls the difference of the weights in the piece B of Proposition VI.7.

The pairs with a common prime $q > z$. Such a pair satisfies $p \equiv -1 \pmod{4u\alpha q}$ with $q \mid u$, and $m = 4u\alpha q \le 4X_j^2 \le 4\cdot 2^{j(1-2\delta)}$. For $x = 2^{j+1}$ we have $\log(x/m) \ge \delta j \log 2$ when $j \ge 3/\delta$. The Brun–Titchmarsh inequality $\pi(x;m,b) \le 2x/(\varphi(m)\log(x/m))$ [20] gives

$$\pi(x;m,-1) \le \frac{2\cdot 2^{j+1}}{\varphi(m)\,\delta\, j\log 2} \le \frac{5\,\pi(I_j)}{\delta\,\varphi(m)}$$

for j large, since $\pi(I_j) \sim 2^j/(j\log 2)$. Summing over $q > z$ and the pairs, with $\varphi(4u\alpha q) = q\,\varphi(4u\alpha)$, the total over $p \in I_j$ is at most

$$\frac{5\,\pi(I_j)}{\delta} \sum_{q>z} \frac{1}{q(q-1)} \sum_{u'\alpha\le X_j} \frac{1}{\varphi(4u'\alpha)} \ll \frac{\pi(I_j)\log^2 X_j}{\delta z}.$$

With the weight $1/p \le 2^{-j}$ this contributes $O(j/(\delta z))$ to $\sum_{p\in I_j} |B_p|/p$.

7. Large common prime factors in the strips

In the pieces B of Propositions VI.7 and VII.7 the pairs (u,α) satisfy $u\alpha \le x^{2/3}$. A pair whose u and a share a prime $q > z$ satisfies $p \equiv -1 \pmod m$ with $m = 4u\alpha q$, $q \mid u$, and we write $u = qu'$. If $q \le x^{1/6}$, then $m \le 4x^{5/6}$, so $\log(2x/m) \ge \frac{1}{7}\log x$ for j large, and the Brun–Titchmarsh inequality gives $\pi(2x;m,-1) \ll x/(\varphi(m)\log x)$. As in Appendix A 6, with $\varphi(4u\alpha q) \ge q(q-1)\varphi(4u'\alpha)$, the sum over $q > z$ and the pairs is

$$\ll \frac{x}{\log x} \sum_{q>z} \frac{1}{q(q-1)} \sum_{u'\alpha\le x^{2/3}} \frac{1}{\varphi(4u'\alpha)} \ll \frac{x\log x}{z},$$

and with the weight $1/p \le 1/x$ it contributes $O(j/z)$. If $q > x^{1/6}$, we use $\pi(2x;m,-1) \le 2x/m + 1$. The first term gives

$$\sum_{q>x^{1/6}} \frac{2x}{4q^2} \sum_{u'\alpha\le x^{2/3}} \frac{1}{u'\alpha} \ll x^{5/6}(\log x)^2,$$

and the second term is at most the number of triples (q,u',α) with $qu'\alpha \le x^{2/3}$, which is $O(x^{2/3}(\log x)^2)$. With the weight $1/p \le 1/x$ both are $o(1)$.[17]

8. The density of the pairs that fail

Profiles. Fix $\xi > 0$. For $x > 1$ and $n \ge 1$, the ξ-profile of n is the pair $\Pi_x(n) = (t, Z)$, where $t = \log T(n)/\log x$, with $T(n)$ the product of the prime powers of n whose prime is below x^ξ, and Z is the multiset of the numbers $\log q/\log x$ over the prime factors $q \ge x^\xi$ of n, counted with multiplicity. The size of n is $s = t + \sum Z$. On the space of profiles let

$$\lambda_1 = \rho(t/\xi)\,dt \otimes \nu, \qquad \nu = \delta_\varnothing + \sum_{k\ge1} \frac{1}{k!} \prod_{i=1}^{k} \frac{\mathbf{1}[z_i \ge \xi]\,dz_i}{z_i},$$

where ν is a measure on finite multisets $\{z_1,\dots,z_k\}$. Every n factors uniquely as $n = T(n)\,r$ with r free of primes below x^ξ, and $1/n = (1/T(n))(1/r)$. By the theorem of Dickman and de Bruijn [26, Chapter III.5], $\Psi(X, X^{1/v}) = X\rho(v)(1+o(1))$ uniformly for bounded v, and partial summation gives

$$\sum_{\substack{T \le x^t \\ T\ x^\xi\text{-smooth}}} \frac{1}{T} = \log x \int_0^t \rho\Big(\frac{\tau}{\xi}\Big)\,d\tau + o(\log x)$$

uniformly for bounded t. By Mertens' theorem,

$$\sum_{x^z < q \le x^{z'}} \frac{1}{q} = \log\frac{z'}{z} + o(1) \qquad (\xi \le z < z'),$$

and the numbers r with a repeated prime factor $q \ge x^\xi$ have total weight $O(x^{-\xi})$ relative to the others. Hence, for every bounded set F of profiles whose boundary is λ_1-null,

$$\frac{1}{\log x} \sum_{n:\ \Pi_x(n)\in F} \frac{1}{n} \to \lambda_1(F).$$

Since the weight $1/(uv)$ is a product, the same holds for pairs: $(\log x)^{-2}\sum_{(\Pi_x(u),\Pi_x(v))\in F} 1/(uv) \to \lambda_1\otimes\lambda_1(F)$ for every bounded F with $\lambda_1\otimes\lambda_1$-null boundary. The image of λ_1 under the size s is Lebesgue measure, since $\sum_{n\le x^s} 1/n = s\log x + O(1)$.

The failure set. Let $\xi = \min(\theta^+ - \theta^-, \theta^-)$, so that $G = x^\xi$ for the window $(x^{\theta^-}, x^{\theta^+})$. The pair (u,v) fails the rule exactly when its profiles (t_1,Z_1), (t_2,Z_2) satisfy: $s_1, s_2 \notin (\theta^-,\theta^+)$; no element of $Z_1\cup Z_2$ lies in (θ^-,θ^+); and the largest element of $(Z_1\cup Z_2)\cap[\xi,\theta^-]$, or 0 if there is none, plus t_1+t_2 is at most θ^-. When $s_1, s_2 > \theta^-$, the last two conditions say that $t_1 + t_2 \le \theta^-$ and that every element of $Z_1\cup Z_2$ lies in $\mathscr{A}_w$ with $w = \max(\theta^- - t_1 - t_2, \xi)$, up to the λ_1-null event that an element equals ξ. The boundary of the failure set lies in the union of the sets where s_i, an element of $Z_1\cup Z_2$, or the sum in the last condition takes one of the values ξ, θ^-, θ^+, which are $\lambda_1\otimes\lambda_1$-null on $\{s_1, s_2 > \theta^-\}$. For fixed t_i, the density of $\sum Z_i$ over the multisets in $\mathscr{A}_w$ is ϖ_w: writing f for this density, $rf(r)$ is the ν-integral of $\sum_i z_i$ over the multisets with sum r, and by the symmetry of ν this equals $\mathbf{1}_{\mathscr{A}_w}(r) + \int_{\mathscr{A}_w\cap(0,r)} f(r-z)\,dz$, which is (25). The empty multiset would give $s_i = t_i \le \theta^-$, which is excluded. Hence the $\lambda_1\otimes\lambda_1$-measure of the failing profiles with $(s_1,s_2)\in R$ is $\iint_R U$.

Lemma A.1. *Let $c > 0$, and let $\mathscr{A} \subset \mathscr{B} \subset (c,\infty)$ be finite unions of intervals. For $\mathscr{S} \in \{\mathscr{A},\mathscr{B}\}$ let $\varpi_\mathscr{S}$ be the function that vanishes on $(-\infty, c]$ and satisfies (25) with $\mathscr{S}$ in place of $\mathscr{A}_w$. Then, for $r > c$,*

$$0 \le \varpi_\mathscr{A}(r) \le \varpi_\mathscr{B}(r) \le \frac{1}{c}\,\omega\Big(\frac{r}{c}\Big) \le \frac{1}{c},$$

where ω is Buchstab's function: $\omega(u) = 1/u$ for $1 \le u \le 2$ and $(u\,\omega(u))' = \omega(u-1)$ for $u > 2$.

Proof. Put $g_\mathscr{S}(z) = \mathbf{1}_\mathscr{S}(z)/z$, let $g_\mathscr{S}^{*k}$ be its k-fold convolution, with $g_\mathscr{S}^{*0}$ the unit mass at 0, and put $f_\mathscr{S} = \sum_{k\ge1} g_\mathscr{S}^{*k}/k!$. Since $g_\mathscr{S}$ vanishes on $(-\infty,c]$, $g_\mathscr{S}^{*k}$ vanishes on $(-\infty,kc]$, and the sum is finite at every point. On the set $z_1+\cdots+z_k = r$ we have $r = z_1+\cdots+z_k$, and the k terms contribute equally, so

$$\begin{aligned} r\,g_\mathscr{S}^{*k}(r) &= k\int z\,g_\mathscr{S}(z)\,g_\mathscr{S}^{*(k-1)}(r-z)\,dz \\ &= k\int_\mathscr{S} g_\mathscr{S}^{*(k-1)}(r-z)\,dz. \end{aligned}$$

Dividing by $k!$ and summing over $k \ge 1$ gives $rf_\mathscr{S}(r) = \mathbf{1}_\mathscr{S}(r) + \int_{\mathscr{S}\cap(0,r)} f_\mathscr{S}(r-z)\,dz$. This equation determines its solution on $(c,2c]$, then on $(2c,3c]$, and so on, so $\varpi_\mathscr{S} = f_\mathscr{S}$. Every term of the series is nonnegative and does not decrease when $\mathscr{S}$ grows, which gives the first two inequalities. For $\mathscr{S} = (c,\infty)$ put $F(u) = c\,f_\mathscr{S}(cu)$; the equation becomes $uF(u) = 1 + \int_1^{u-1} F(v)\,dv$ for $u > 1$, which is the integral form of Buchstab's equation, so $F = \omega$. Finally $\omega \le 1$: this holds on $[1,2]$, and if it holds on $[1,u-1]$ then $u\,\omega(u) = 1 + \int_1^{u-1}\omega \le u-1$. □

The bound $0 \le U \le 1$. Let ϖ be the solution of (25) with $\mathscr{A}_w$ replaced by (ξ,∞), the density of $\sum Z$ over all multisets. Then $0 \le \varpi_w \le \varpi$ by Lemma A.1, and dropping the condition $t_1+t_2 \le \theta^-$ in (26),

$$U(s_1,s_2) \le \prod_{i=1}^{2} \int_0^{s_i} \rho\Big(\frac{t}{\xi}\Big)\varpi(s_i - t)\,dt \le 1,$$

because each factor is the λ_1-density at s_i of the profiles with a nonempty multiset, and the total density is 1.

Proof of Lemma VI.4. For $h = \mathbf{1}_{(1,1)}$ the statement is the convergence above, applied to the passing profiles with

[17] The two ranges of q are needed because the Brun–Titchmarsh inequality loses its strength when the modulus m approaches x: for $q \le x^{1/6}$ the modulus is at most $x^{5/6}$ and the inequality saves the factor $\log x$, while for larger q the square q^2 in the denominator is already small enough. The bound $x^{2/3}$ for $u\alpha$ holds on the strips, where $u\alpha \le x^{1/2+1/42}$, and on the deep strips, where the sizes satisfy $2s_1+s_2<1$ and $s_1+2s_2<1$.

$(s_1,s_2)\in R$, which form a bounded set with null boundary and have measure $\iint_R(1-U)$. For general h,

$$\sum_{(u,v)}\frac{W(u,v)}{uv}=\sum_{d,e}\frac{h(d,e)}{de}\sum_{(u',v')}\frac{1}{u'v'},$$

where (u',v') runs over the pairs for which (du',ev') passes and has its sizes in R. For fixed d,e and x^{ξ} larger than every prime factor of de, the profiles of du' and ev' are those of u' and v' with t increased by $\log d/\log x$ and $\log e/\log x$, which tend to 0; since the limit set has null boundary, the inner sum is $(\log x)^2\iint_R(1-U)+o((\log x)^2)$. It is also at most $\sum_{u',v'\le x^K}1/(u'v')\ll(\log x)^2$, so dominated convergence over (d,e) with $\sum|h(d,e)|/(de)<\infty$ gives the lemma.

Proof of the limit in Lemma VII.10. Fix $\eta_0>0$. For $\vartheta\in[\eta_0,\frac{1}{42}-\eta_0]$ the number $\xi(\vartheta)=\min(\theta(\vartheta)-2\vartheta,2\vartheta)$ is at least $\xi_0=\min(\frac{2}{105},2\eta_0)$, and the same holds for the windows (θ_i^-,θ_i^+) with $\vartheta_i\in[\eta_0,\frac{1}{42}-\eta_0]$ once δ is small. The ξ-profile of a number is a function of its ξ_0-profile, so all these failure sets live in the space of ξ_0-profiles with the measure $\lambda_1\otimes\lambda_1$ at level ξ_0, and $\sum_i\iint_{R_i}U_i$, restricted to these depths, is the measure of the set of profile pairs with $\vartheta=s_1+s_2-\frac12$ in the range and (s_1,s_2) in $\bigcup_i R_i$ that fail the rule for the window of their piece. As $\delta\to0$ these windows converge to $(2\vartheta,\theta(\vartheta))$, and outside a null set of profiles the indicator of failure converges to that for the limiting window. The sets lie in a fixed bounded region of finite measure, so by dominated convergence the measure tends to $\int_{\eta_0}^{1/42-\eta_0}\ell(\vartheta)\,d\vartheta$. The depths outside $[\eta_0,\frac{1}{42}-\eta_0]$ contribute $O(\eta_0)$ to both sides, because $0\le U\le1$, and η_0 is arbitrary.

9. Splitting the moduli by boxes

For a multiset B of boxes let $\mathscr{S}_B$ be the set of squarefree integers r whose prime factors lie above x^{ε_0} and whose multiset of boxes is B, and let m_k be the multiplicity of the box k in B. The nominal size of B is $\sigma_B=\sum_{k\in B}kh$.

Lemma A.2. *Let B be a multiset of boxes and $G\in(0,\frac{1}{10})$, and suppose that the nominal profile*

$$\Big(2\varepsilon_1+\sum_{\substack{k\in B\\ kh<G}}kh,\ \{kh:k\in B,\ kh\ge G\}\Big),$$

of size $s'=\sigma_B+2\varepsilon_1$, is factorable with margin μ for G, where $2\varepsilon_1<G$. Let ϖ be a function supported on the x^{ε_0}-smooth integers below $x^{2\varepsilon_1+o(1)}$ with $|\varpi|\le\tau_3$, and put

$$\lambda_q=\sum_{q=4Dr}\varpi(D)\,\mathbf{1}_{\mathscr{S}_B}(r).$$

Then for every $n\in[\frac23-s'-\mu,\frac13+\mu]$ there are an integer $C\ge1$ and sequences $\gamma_1,\gamma_2,\gamma_3$ bounded by τ^2 and supported on $[1,x^{s_i+o(1)}]$, with (s_1,s_2,s_3) admissible for n with margin μ, such that $\lambda-C^{-1}\gamma_1*\gamma_2*\gamma_3$ is bounded by τ^6 and supported on the integers divisible by the square of a prime above x^{ε_0}.

Proof. Let $(Z_1,Z_2,Z_3,L,F,(x_i))$ be an assignment that serves n with margin μ. The elements of Z are the numbers $kh\ge G$ with $k\in B$, so the assignment places these boxes into three groups. The remaining items are the dust, of nominal size $2\varepsilon_1$, followed by the boxes $k\in B$ with $kh<G$ in increasing order; all of them have nominal size below G. Go through the groups $i\in F$ in increasing order of i and add items, in their order, to the group i until its nominal size reaches x_i; it then lies in $[x_i,x_i+G)$. Before the last group of F is filled, the items used have total nominal size less than $\sum_{i\in F}(x_i-b_i)+(|F|-1)G-(x_{i_1}-b_{i_1})$, where i_1 is that group, so by (28) enough items remain to fill it. The items left over go to the group L. This gives a partition $B=B_1\sqcup B_2\sqcup B_3$ and a group i_0 that holds the dust, and the nominal sizes s_i of the groups (with $2\varepsilon_1$ added for i_0) satisfy $s_i\in[x_i,x_i+G)$ for $i\in F$, $s_i=b_i$ for $i\notin F\cup\{L\}$ and $s_L=s'-\sum_{i\ne L}s_i$. So (s_1,s_2,s_3) is one of the triples of Definition VII.2 and is admissible for n with margin μ.

Put $\gamma_i=\mathbf{1}_{\mathscr{S}_{B_i}}$ for $i\ne i_0$ and $\gamma_{i_0}(m)=\sum_{m=4Dr}\varpi(D)\mathbf{1}_{\mathscr{S}_{B_{i_0}}}(r)$. A prime with box k is at most x^{kh}, so γ_i is supported on $[1,x^{s_i+o(1)}]$, and $|\gamma_{i_0}|\le\tau_3\le\tau^2$. Since the prime factors of D lie below x^{ε_0} and those of r above it, an integer $q=4Dr$ determines D and r. If $r\in\mathscr{S}_B$, the number of factorisations $r=r_1r_2r_3$ with $r_i\in\mathscr{S}_{B_i}$ is

$$C=\prod_k\frac{m_k!}{m_{k,1}!\,m_{k,2}!\,m_{k,3}!},$$

where $m_{k,i}$ is the multiplicity of k in B_i, so $\gamma_1*\gamma_2*\gamma_3(q)=C\lambda_q$. If $\gamma_1*\gamma_2*\gamma_3(q)\ne0$ and $r\notin\mathscr{S}_B$, then $r=r_1r_2r_3$ with $r_i\in\mathscr{S}_{B_i}$ is not squarefree, so the square of a prime above x^{ε_0} divides q. Finally $\gamma_1*\gamma_2*\gamma_3\le\tau_3\cdot\tau^2\le\tau^4$ and $|\lambda|\le\tau^2$. □

The weight λ^β of Proposition VII.7 has this form with $B=\beta_1\sqcup\beta_2$, multiplied by $c_\beta=\prod_k\binom{m_k}{m_{k,1}}$, where $m_{k,1}$ is the multiplicity of k in β_1: an integer $r\in\mathscr{S}_B$ splits in c_β ways as $r(u)r(a)$ with $\beta(u)=\beta_1$ and $\beta(a)=\beta_2$, and $\varpi(D)$ is the number of ways to write $D=D(u)D(a)$ with the conditions of the piece.

Lemma A.3. *Let $\mathscr{G}$, μ and ε_1 be as in Definition VII.5, with $\varepsilon_1\le\mu/48$ and $\varepsilon_1^3\le\min\mathscr{G}$. Let (u,v) be a pair whose dust is at most x^{ε_1}, and $G\in\mathscr{G}$. If the combined G-profile of (u,v) is factorable with margin 2μ and no prime factor of uv has size in $[G-h,G)$, then the nominal profile of $(\beta(u),\beta(v))$ for G is factorable with margin μ.*

Proof. Let (t,Z) be the combined profile, of size s, and (t',Z') the nominal one, of size s'. A prime of size $z\ge G$ has nominal size in $[z,z+h)$, and a prime of size below $G-h$ has nominal size below G, so the primes counted in Z' are those counted in Z; their sizes are at least $G+2\mu$, so no element of Z' lies within μ of G. The number t' is $2\varepsilon_1$ plus the nominal sizes of the primes above x^{ε_0} in t, so $t'\ge t$. There are at most $1/\varepsilon_0$ prime factors above x^{ε_0} and at most $1/G$ above x^G, so $0\le s'-s\le2\varepsilon_1+h/\varepsilon_0+h/G\le4\varepsilon_1$.

Let $n\in[\frac23-s'-\mu,\frac13+\mu]$; then $n\in[\frac23-s-2\mu,\frac13+2\mu]$, and some assignment $(Z_i,L,F,(x_i))$ serves n for (t,Z) with margin 2μ. Take the same partition of the primes, the same L and F, and the targets $x_i'=x_i+b_i'-b_i$, where $b_i'\ge b_i$ are

the nominal and actual sums of the groups; $b_i' - b_i \le h/G \le \varepsilon_1$. The left side of (28) is unchanged and $t' - \mu \ge t - 2\mu$, so the budget holds with margin μ. A triple d' for the new assignment has $d_i' \in [x_i', x_i' + G]$ for $i \in F$, $d_i' = b_i'$ for $i \notin F \cup \{L\}$ and $d_L' = s' - \sum_{i \ne L} d_i'$. Put $d_i = d_i' - (b_i' - b_i)$ for $i \ne L$ and $d_L = s - \sum_{i \ne L} d_i$; this is a triple for the old assignment, so it is admissible for n with margin 2μ, and $|d_i' - d_i| \le \varepsilon_1$ for $i \ne L$, $|d_L' - d_L| \le |s' - s| + 2\varepsilon_1 \le 6\varepsilon_1$. In each of the four conditions (27) the coefficients of d_1, d_2, d_3 have absolute values adding up to at most 8, so each left side changes by at most $48\varepsilon_1 \le \mu$. □

Lemma A.4. *Lemma VI.4 holds with the passing pairs replaced by the pairs (u,v) whose pair of ξ-profiles lies in a bounded set F with $\lambda_1 \otimes \lambda_1$-null boundary, and with $\iint_R (1 - U)$ replaced by $\lambda_1 \otimes \lambda_1(F)$.*

Proof. The proof of Lemma VI.4 in Appendix A 8 uses the set of passing profiles only through its boundedness and the nullity of its boundary. □

10. A certified bound for $\mathscr{J}$

This appendix proves Proposition X.2. For $0 < \vartheta < \frac{1}{42}$ write $\alpha = 2\vartheta$, $\theta = \theta(\vartheta)$, $\xi = \min(\theta - \alpha, \alpha)$ and $S = \frac{1}{2} + \vartheta$, and for $t \ge 0$ let ϖ_t be the function ϖ_w of (25) for the window (α, θ) with $w = \max(\alpha - t, \xi)$.

Lemma A.5. *For $0 < \vartheta < \frac{1}{42}$,*

$$\ell(\vartheta) \le \int_0^{\alpha} P(t)\, Q(t)\, dt,$$

where

$$P(t) = \int_0^t \rho\Big(\frac{t_1}{\xi}\Big)\rho\Big(\frac{t - t_1}{\xi}\Big) dt_1,$$
$$Q(t) = \int_{\theta - t}^{S - \theta} \varpi_t(r)\, \varpi_t(S - t - r)\, dr.$$

Proof. Since $\theta < \frac{1}{4}$, the two intervals (α, θ) and $(S - \theta, S - \alpha)$ are disjoint, and $U_\vartheta(s, S - s) = 0$ when s lies in either of them. For the other s in $(\alpha, S - \alpha)$, that is for $\theta \le s \le S - \theta$, $U_\vartheta(s, S - s)$ is the integral (26), whose integrand is nonnegative. For fixed $(t_1, t_2) \in \Delta$ put $t = t_1 + t_2$, so that $w = \max(\alpha - t, \xi)$, and substitute $r = s - t_1$. Then $S - s - t_2 = S - t - r$, and r runs over $[\theta - t_1, S - \theta - t_1] \subset [\theta - t, S - \theta]$. Hence the integral over s is at most $Q(t)$, and integrating over Δ along the segments $t_1 + t_2 = t$ gives the lemma.[18] □

[18] The only loss in Lemma A.5 is the widening of the range of r from $[\theta - t_1, S - \theta - t_1]$ to $[\theta - t, S - \theta]$. The condition that s avoids the window is kept, and it matters: dropping it, that is integrating r over $[\alpha - t, S - \alpha]$, raises the computed bound by about 6%.

Bounds on a cell. Fix $0 < a < b < \frac{1}{42}$. The function θ is decreasing and concave, $\theta - 2\vartheta$ is decreasing, and ξ is concave as a minimum of linear functions. Hence, for every $\vartheta \in [a, b]$,

$$\xi_- \le \xi \le \xi_+, \qquad \theta \ge \theta(b), \qquad 2a \le \alpha \le 2b,$$
$$\tfrac{1}{2} + a \le S \le \tfrac{1}{2} + b,$$

with $\xi_- = \min(\xi(a), \xi(b))$ and $\xi_+ = \min(2b, \theta(a) - 2a)$. Let $0 = \tau_0 < \tau_1 < \cdots < \tau_K$. For $\tau_k \le t \le \tau_{k+1}$ the set $\mathscr{A}_w$ of (25), with $w = \max(\alpha - t, \xi)$, lies in

$$\mathscr{A}_k = \big(\xi_-, \max(2b - \tau_k, \xi_+)\big] \cup [\theta(b), \infty),$$

so $\varpi_t \le \varpi_{\mathscr{A}_k}$ by Lemma A.1. Since ρ is nonincreasing, $\rho(t_1/\xi) \le \rho(t_1/\xi_+)$. Let P_+ be the function P of Lemma A.5 with ξ_+ in place of ξ. If $\tau_K \ge 2b$, then $\tau_K \ge \alpha$, and Lemma A.5 gives, for every $\vartheta \in [a, b]$,

$$\ell(\vartheta) \le \sum_{k < K} Q_k \int_{\tau_k}^{\tau_{k+1}} P_+(t)\, dt, \tag{A3}$$

where

$$Q_k = \int_{\theta(b) - \tau_{k+1}}^{1/2 + b - \theta(b)} \varpi_{\mathscr{A}_k}(r) \sup_{y \in Y_k(r)} \varpi_{\mathscr{A}_k}(y)\, dr$$

and $Y_k(r) = [\frac{1}{2} + a - \tau_{k+1} - r, \frac{1}{2} + b - \tau_k - r]$. If instead $24\xi_+ < 2b$ and $\tau_K \ge 24\xi_+$, the right side of (A3) gets the extra term

$$E = \frac{13}{6}\, e^{\gamma} \rho(12)\, \xi_+^2 \cdot \frac{\frac{1}{2} + b}{\xi_-^2} \tag{A4}$$

for the part $t > 24\xi_+$. Indeed, $\varpi_t \le 1/\xi_-$ by Lemma A.1, so $Q(t) \le (\frac{1}{2} + b)/\xi_-^2$; and

$$\int_{24\xi_+}^{\infty} P_+(t)\, dt \le 2 \int_{12\xi_+}^{\infty} \rho\Big(\frac{t_1}{\xi_+}\Big) dt_1 \int_0^{\infty} \rho\Big(\frac{t_2}{\xi_+}\Big) dt_2$$
$$= 2\xi_+^2 e^{\gamma} \int_{12}^{\infty} \rho,$$

because $t_1 + t_2 > 24\xi_+$ forces $t_1 > 12\xi_+$ or $t_2 > 12\xi_+$, and $\int_0^{\infty} \rho = e^{\gamma}$ [26, Chapter III.5]. From $v\rho(v) = \int_{v-1}^{v} \rho \le \rho(v - 1)$ we get $\rho(v) \le \rho(v - 1)/13$ for $v \ge 13$, so $\int_{12}^{\infty} \rho \le \rho(12) \sum_{k \ge 0} 13^{-k} = \frac{13}{12}\rho(12)$. Integrating (A3) over the cell gives $\int_a^b \ell \le (b - a)\, \bar{\ell}$, where $\bar{\ell}$ is its right side. Proposition X.1 also gives $\ell(\vartheta) \le \frac{1}{2} + b - 2\theta(b)$ on the cell, and we use the smaller of the two bounds.

Dickman's function on a grid. Let $h = 2^{-10}$, and define $r_i = 1$ for $0 \le i \le 2^{10}$ and

$$r_i = \frac{1}{i} \sum_{j = i - 2^{10}}^{i - 1} r_j \qquad (i > 2^{10}).$$

Since $u\rho(u) = \int_{u-1}^{u} \rho$ and ρ is nonincreasing, $ih\,\rho(ih) \le h \sum_{j = i - 2^{10}}^{i - 1} \rho(jh)$, and induction on i gives $\rho(ih) \le r_i$. Let $g \ge \xi_+ h$. The band $mg \le t_1 + t_2 \le (m + 1)g$ of the quadrant is covered by the lower halves of the squares $[ig, (i + 1)g] \times [jg, (j + 1)g]$ with $i + j = m$ and the upper halves of those

with $i+j=m-1$, each of area $g^2/2$, and on such a square $\rho(t_1/\xi_+)\rho(t_2/\xi_+) \le r_i r_j$. Hence

$$\int_{mg}^{(m+1)g} P_+(t)\,dt \le \frac{g^2}{2}\,(c_m + c_{m-1}), \qquad c_m = \sum_{i+j=m} r_i r_j,$$

with $c_{-1}=0$. We take g to be $\xi_+ h$ rounded up and $\tau_k = 32kg$, and stop at the first K with $\tau_K \ge \min(2b, 24\xi_+)$.

The functions $\varpi_{\mathscr{A}_k}$ on a grid. Let $\mathscr{A} = (x_0, w] \cup [t_0, \infty)$ with $0 < x_0 \le w < t_0$, and let η be a power of 2 with $\eta \le x_0/16$. Put $p_j = 0$ if $(j+1)\eta \le x_0$, and otherwise

$$p_j = \frac{e_j + \eta \sum_{m \in M_j} p_m}{\max(j\eta, x_0)},$$

where $e_j = 0$ if $w < j\eta$ and $(j+1)\eta < t_0$, and $e_j = 1$ otherwise, and M_j is the list, with repetitions, of the indices m whose interval $[m\eta, (m+1)\eta]$ meets $[j\eta - w, (j+1)\eta - x_0]$, followed by those whose interval meets $(0, (j+1)\eta - t_0]$. Every $m \in M_j$ satisfies $m < j$, because $\eta < x_0$, so the p_j are defined in turn. For r in $[j\eta, (j+1)\eta]$ with $r > x_0$, the right side of (25) is at most e_j plus the integral of $\varpi_{\mathscr{A}}$ over $[j\eta - w, (j+1)\eta - x_0]$ and over $(0, (j+1)\eta - t_0]$, since $r - z$ lies in the first interval for $z \in (x_0, w]$ and in the second for $z \in [t_0, r)$. Induction on j therefore gives $\varpi_{\mathscr{A}} \le p_j$ on $[j\eta, (j+1)\eta]$. We take for η the largest power of 2 not above $\xi_-/64$, and bound Q_k by the sum, over the intervals $[m\eta, (m+1)\eta]$ that meet the range of r, of ηp_m times the largest p_q over the intervals $[q\eta, (q+1)\eta]$ that meet $Y_k(r)$ for some r in $[m\eta, (m+1)\eta]$.

The computation. On $[0, 2^{-18}]$ and on $[\frac{1}{42} - 10^{-6}, \frac{1}{42}]$ we use the bound $\ell(\vartheta) \le \frac{1}{2} + \vartheta - 2\theta(\vartheta)$ of Proposition X.1, which is nondecreasing in ϑ, at the right end. The rest of $[0, \frac{1}{42}]$ is cut into cells: from 2^{-18} to 0.005 the cells grow by the factor 1.05; $[0.005, \frac{1}{54}]$ and $[\frac{1}{54}, \frac{1}{44}]$ are cut into 120 equal cells each; and from $\frac{1}{44}$ to $\frac{1}{42} - 10^{-6}$ the distance to $\frac{1}{42}$ shrinks by the factor 1.03 from cell to cell. On each cell the bound $(b-a)\bar{\ell}$ is evaluated as above. The program sets the rounding mode of the processor to rounding upward, so that every sum, product and quotient it computes is at least the exact value; it obtains lower bounds, such as ξ_- and $\theta(b)$, as the negatives of upper bounds of the negatives. Every grid point is an exact multiple of a power of 2, consecutive cells share their end points exactly, and the bounds of the cells are added in the same rounding mode. Table IV shows the result.

TABLE IV. The certified bound for $\mathscr{J} = \int_0^{1/42} \ell$ by ranges of the depth ϑ, in units of 10^{-6} and rounded up, and the value of the same integral by the quadrature of Section X. The first and last ranges include $[0, 2^{-18}]$ and $[\frac{1}{42} - 10^{-6}, \frac{1}{42}]$, where the bound of Proposition X.1 replaces the cells.

Range of ϑ	Cells	Bound	Quadrature
$[0, 0.005]$	148	2.37	1.03
$[0.005, \frac{1}{54}]$	120	81.22	75.76
$[\frac{1}{54}, \frac{1}{44}]$	120	94.19	87.50
$[\frac{1}{44}, \frac{1}{42}]$	237	156.36	141.44
Total	625	334.13	305.73

11. A certified bound for $\mathscr{K}$

This appendix proves Proposition X.3. In the coordinates (s_1, ϑ),

$$\mathscr{K} = \int_{1/42}^{1/6} k(\vartheta)\,d\vartheta,$$

where $k(\vartheta)$ is the $\lambda_1 \otimes \lambda_1$-density, at the size $s = \frac{1}{2} + \vartheta$ of the product, of the factorable pairs with $2\vartheta < s_1 < \frac{1}{2} - \vartheta$. Fix a cell $[a, b]$ of depths and a number G. We bound $k(\vartheta)$ from below on the whole cell by the density of the pairs whose combined G-profile is factorable.

Configurations. The combined G-profile of a pair is (R, Z), where $R = t_1 + t_2$ is the total size of the prime factors below x^G and Z the multiset of the sizes of the others. Let z_1 be the largest element of Z and $Z^\flat$ the others. A configuration is a list of intervals J with multiplicities m_J for the elements of $Z^\flat$, together with an interval for R; given s, the largest element is $z_1 = s - \sum Z^\flat - R$, and the configuration is valid if z_1 exceeds G and the right end of every interval. By Appendix A 8, the density at s of the pairs whose combined profile lies in a valid configuration is

$$\prod_J \frac{1}{m_J!} \Big(\int_J \frac{dz}{z} \Big)^{m_J} \sum_{\text{labels}} \int \frac{\Psi_c(R)}{z_1}\,dR,$$

where a label assigns each element of Z to u or to v, and $\Psi_c(R) = \int \rho(t/G)\rho((R-t)/G)\,dt$ over the $t = t_1$ for which the label meets the conditions $s_1, s_2 > 2\vartheta$. The program bounds this from below on the cell: it uses $2b$ in place of 2ϑ in the conditions on s_1, s_2, the band tables of Dickman's function of Appendix A 10 for Ψ_c, and Jensen's inequality for $1/z_1$, with every operation rounded in the direction of the bound.

The test. A configuration passes if for every profile in it and every $n \in [\frac{1}{6} - b, \frac{1}{3}] \supset [\frac{2}{3} - s, \frac{1}{3}]$ some assignment serves n with margin 0; by Lemma VII.3 its profiles are then factorable, except for the null set with an element equal to G. The program places the intervals of the configuration into the three groups in all ways, chooses L, the set F and targets x_i, and finds by linear programming an interval $[n_-, n_+]$ of n with targets at its two ends. The conditions (27) and (28), for fixed elements of Z, fixed R and fixed positions of the filled sizes in $[x_i, x_i + G]$, are affine in (x_i, n), so the set of (x_i, n) for which they hold is convex, and it is enough to check the two ends. At each end the program checks them for every configuration in the box with rounding in the direction of failure, by taking each variable at the end of its interval that is worse for the inequality. The configuration passes if these intervals of n cover $[\frac{1}{6} - b, \frac{1}{3}]$.

The tree. For each number $k \le k_{\max}$ of prime factors above x^G the program starts from the configuration with $k-1$ elements in $[G, T]$, $T \ge \frac{2}{3}$, and $R \in [0, 16G]$, and splits a configuration that is not valid, or valid but neither passing nor failing at the smallest width, by halving the interval for R or

an interval of $Z^{\flat}$, with its multiplicity shared out between the two halves. Configurations whose measure is below a threshold are dropped. The bound for the cell is the sum of the lower bounds over the passing configurations; the configurations are disjoint, and those with more than $k_{\max}$ elements, with $R > 16G$, dropped, failing or undecided add nothing.

Parameters. The cells are $[\frac{k}{1024}, \frac{k+1}{1024}]$ with $24 \le k \le 70$, with $G = 0.04$ for $k \le 45$ and $G = 0.03$ for $k \ge 46$, and $k_{\max} = 7$ on every cell. The interval for R is cut into bands of width $G/128$. A configuration that is not decided is halved until its intervals of $Z^{\flat}$ have width 0.00625 and its interval for R four bands; a configuration that is not valid, or that passes with a lower bound more than 5% below the estimate of its measure, is halved down to one band and the same width. The threshold for dropping is 10^{-8}. The cells took 5 hours in all.

Validation. For a cell at $\vartheta \approx 0.03$ the program also draws $1.39 \cdot 10^9$ random profiles from the passing configurations, together with random fill sizes, and checks the four conditions directly; there was no violation.

12. Partial summation in a block

Let $f(p) \ge 0$ and $F(y) = \sum_{2^j < p \le y} f(p)$ for $y \in I_j$. Suppose $F(y) = (\pi(y) - \pi(2^j))S + E(y)$ with $|E(y)| \le E$ for all $y \in I_j$. Then

$$\sum_{p \in I_j} \frac{f(p)}{p} = \frac{F(2^{j+1})}{2^{j+1}} + \int_{2^j}^{2^{j+1}} \frac{F(y)}{y^2}\, dy = S \sum_{p \in I_j} \frac{1}{p} + O\Big(\frac{E}{2^j}\Big),$$

because the same identity holds for $\pi(y) - \pi(2^j)$ in place of F and $\int_{2^j}^{2^{j+1}} y^{-2}\, dy = 2^{-j-1}$. With $E \ll 2^j (\log 2^j)^{-3}$ the error is $O(j^{-3})$. The pairs removed in Appendix A 6 are handled with the trivial weight $1/p \le 2^{-j}$. Finally, the prime number theorem with the error term $O(y \exp(-c\sqrt{\log y}))$ gives $\sum_{p \le y} 1/p = \log\log y + M_1 + O(\exp(-c\sqrt{\log y}))$, so $\sum_{p \in I_j} 1/p = \log(1 + 1/j) + O(e^{-c\sqrt{j}}) = 1/j + O(1/j^2)$.

13. From the density to the Euler product

Let $N(\lambda) = \sum_{p \le e^\lambda} \omega(p)/p$ and $G_0(\sigma) = \sum_p (\omega(p)/p) p^{-\sigma}$. Since $N(\lambda) = O(\lambda^2)$, integration by parts gives

$$G_0(\sigma) = \int_0^\infty e^{-\sigma\lambda}\, dN(\lambda) = \sigma \int_0^\infty e^{-\sigma\lambda} N(\lambda)\, d\lambda.$$

If $N(\lambda) \le (a+\varepsilon)\lambda^2 + C_\varepsilon$, then $G_0(\sigma) \le (a+\varepsilon)\sigma \cdot 2\sigma^{-3} + C_\varepsilon = 2(a+\varepsilon)\sigma^{-2} + C_\varepsilon$. In the same way

$$\sum_p \frac{\omega(p)\log p}{p^{1+\sigma}} = \int_0^\infty N(\lambda)(\sigma\lambda - 1) e^{-\sigma\lambda}\, d\lambda,$$

$$\sum_p \frac{\omega(p)\log^2 p}{p^{1+\sigma}} = \int_0^\infty N(\lambda)(\sigma\lambda^2 - 2\lambda) e^{-\sigma\lambda}\, d\lambda,$$

and, when $N(\lambda) = (a + o(1))\lambda^2$, these are $(4a + o(1))\sigma^{-3}$ and $(12a + o(1))\sigma^{-4}$ as $\sigma \to 0^+$, from $\int_0^\infty \lambda^2 (\sigma\lambda - 1) e^{-\sigma\lambda} d\lambda = 4\sigma^{-3}$ and $\int_0^\infty \lambda^2 (\sigma\lambda^2 - 2\lambda) e^{-\sigma\lambda} d\lambda = 12\sigma^{-4}$. If the sums are restricted to $p \le Q = e^T$, the same asymptotics hold provided $\sigma T \to \infty$, since the tails are bounded by $\sigma^{-k} \int_{\sigma T}^\infty x^{k+1} e^{-x}\, dx$ with $k = 2, 3, 4$. Finally $0 \le g(p) - \omega(p)/p \le 2\omega(p)^2/p^2$ once $\omega(p) \le p/2$, which holds for all large p because $\omega(p) \le C_\eta p^\eta$; hence $F(\sigma) = G_0(\sigma) + O(1)$, and the same holds for the sums with $\log p$ and $\log^2 p$.

14. The choice of σ

Assume $N(\lambda) = (a + o(1))\lambda^2$ with $a > 0$. With $y_p = g(p) p^{-\sigma}$ and $\pi_p = y_p/(1 + y_p)$,

$$\mu_\sigma = \sum_{p \le Q} \pi_p \log p, \qquad V_\sigma = \sum_{p \le Q} \pi_p (1 - \pi_p) \log^2 p.$$

Both are continuous in σ, and both tend to 0 as $\sigma \to \infty$ because the sums are finite. Since $y_p - y_p^2 \le \pi_p \le y_p$, Appendix A 13 gives, whenever $\sigma T \to \infty$,

$$\mu_\sigma = (4a + o(1))\sigma^{-3}, \qquad V_\sigma \le (12a + o(1))\sigma^{-4}.$$

Take $\sigma_1 = \varepsilon_0 T^{-1/3}$ with $\varepsilon_0 > 0$ small. Then $\sigma_1 T \to \infty$ and $\mu_{\sigma_1} \ge 3a\varepsilon_0^{-3} T > T$, so by continuity there is $\sigma \ge \sigma_1$ with $\mu_\sigma + \sqrt{2V_\sigma} = T$. If σT stayed bounded along a sequence of T, say $\sigma T \le K$, then $p^{-\sigma} \ge e^{-K}$ for $p \le Q$ and

$$\mu_\sigma \ge \frac{e^{-K}}{2} \sum_{p \le Q} \frac{\omega(p)\log p}{p} \gg e^{-K} T^3,$$

contradicting $\mu_\sigma \le T$ for large T. Hence $\sigma T \to \infty$, the asymptotics apply, and $\mu_\sigma \le T$ gives $\sigma \ge (4a - o(1))^{1/3} T^{-1/3}$, while $\sigma \ge \sigma_1$ already gives $\sigma \gg T^{-1/3}$ from below and $T = \mu_\sigma + \sqrt{2V_\sigma} \ll \sigma^{-3}$ gives $\sigma \ll T^{-1/3}$. Therefore $\sigma\sqrt{2V_\sigma} \ll \sigma^{-1} \ll T^{1/3}$. Finally

$$\sum_{p \le Q} \log(1 + y_p) \ge \sum_{p \le Q} y_p - \frac{1}{2} \sum_p y_p^2 \ge F_Q(\sigma) - O(1),$$

where F_Q is F restricted to $p \le Q$, and $F_Q(\sigma) = (2a + o(1))\sigma^{-2}$ by Appendix A 13. This is what the proof of Proposition VIII.2(ii) uses.

15. The band below $\sqrt{p}$

Proposition IX.1 uses the following count. For $p \in I_j$ take the pairs (u, α) with $p^{1/2-\delta} < u\alpha \le p^{1/2}$ and $p \equiv -1 \pmod{4u\alpha}$; the corners A and C are treated in the same way. Every such pair has $2^{j(1/2-\delta)} < u\alpha \le 2^{(j+1)/2}$, and the modulus $m = 4u\alpha$ satisfies $\log(2^{j+1}/m) \ge (\frac{j}{2} - 2)\log 2$. The Brun–Titchmarsh inequality gives $\pi(2^{j+1}; m, -1) \le (9 + o(1))\pi(I_j)/\varphi(m)$, so the number of these pairs, summed over $p \in I_j$, is at most

$$(9 + o(1))\, \pi(I_j) \sum_{2^{j(1/2-\delta)} < n \le 2^{(j+1)/2}} \frac{\tau(n)}{\varphi(4n)}.$$

The sum over n is $O(\log^2(2^{j/2}) - \log^2(2^{j(1/2-\delta)})) + O(j) = O(\delta j^2) + O(j)$. With the weight $1/p \le 2^{-j}$, the contribution of the block is $O(\delta j) + O(1)$, and the sum over $j \le J$ is $O(\delta J^2) + O(J) = O(\delta\lambda^2) + O(\lambda)$.

DATA AVAILABILITY

The LaTeX source of this paper, the Lean 4 formalisation and the C, C++, Python, Julia and shell programs described in Section XII, the records of the three engine runs and the scripts that reproduce them are available at `https://github.com/creelie/ErdosProblem1` and in the archive [28]. The corners, the treatment of the centre and the rule of Definition VI.2 were selected with the engine Anaxia of the BLLM project [1, 3], version 3.0.5, in the first two runs described in Section XII, and the splitting of the moduli in the deep strips with version 3.1.0 [2] in the third. Deep Bhattacharjee carried out the computations and the Lean, C, C++, Python, Julia and shell verifications, and the proofs and the text were drafted, with AI assistance (Claude, Anthropic); both authors approved the final version.

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